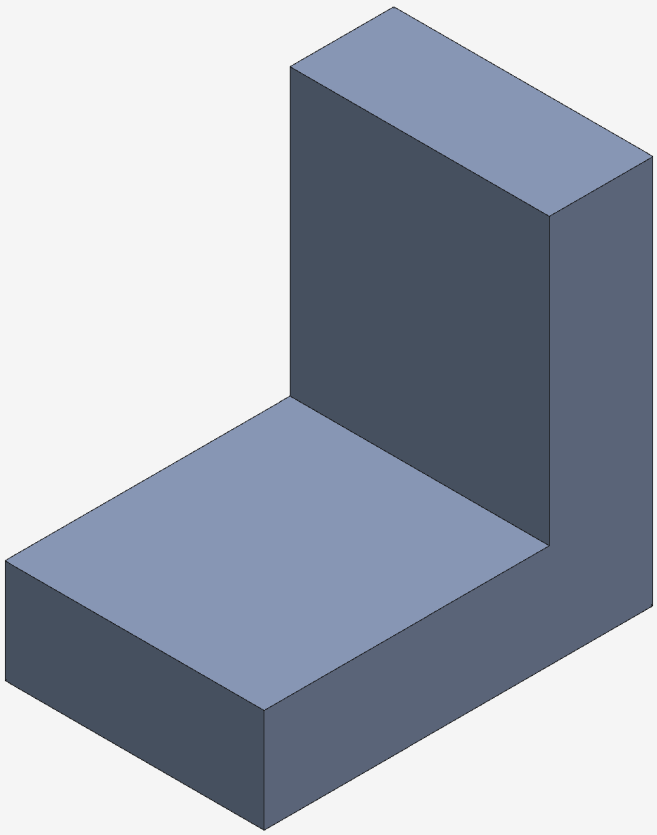


USER MANUAL

Saulitwerx

Open-source parametric 3D CAD

Describes	`main` at commit `1a8c4c2`
Built	2026-11-04
Status	Pre-alpha: no desktop app yet

How to use this manual

This manual teaches you to use Saulitwerx, an open-source program for designing mechanical parts on a computer. You don't need to know anything about computer-aided design (CAD) to read it. Where a word has a special meaning, the manual explains it the first time it appears, and again in the glossary at the back.

Saulitwerx is young

Saulitwerx is being built in the open, one small step each night. Today it is **pre-alpha**: the parts that do the real geometry work exist and are tested, but there is no desktop app with windows and buttons yet. That changes what this edition of the manual can offer:

- The **Concepts** part is for everyone. It explains how parametric CAD works, with pictures, so you will be ready when the app arrives.
- The **Tasks** part shows what you can already do, by running real commands and small programs. These sections carry a purple **For developers** band, because they need a terminal and some comfort with installing software.
- Nothing in this manual describes a feature as working unless it works on the version named on the cover. Plans are kept apart, in **Coming next** at the back.

How the book is organized

Part	Read it when you want to
I. Getting started	Know what Saulitwerx is, install it, and make your first picture of a part.
II. Concepts	Understand sketches, constraints, dimensions, parameters, features and solids.
III. Tasks	Do one specific thing. Each section is one task, with numbered steps.
IV. Reference	Look up an exact option, default or allowed value.
V. Help	Fix a problem, or understand an error message.
Back matter	Look up a word, see what changed, or find a topic in the index.

Conventions

You see	It means
`swx-qa render`	Text you type exactly, or text the program prints.
Enter	A key on your keyboard.
sketch	A term defined at this spot. It is also in the glossary and the index.
1	A numbered step. Steps start with what to do, then say what you should see.
RESULT	What you have when a task is done.

> **TIP**
>
> Callouts like this one hold shortcuts and good habits.

> **CAUTION**
>
> Cautions warn about a mistake that is easy to make, and how to avoid it.

> **FOR DEVELOPERS**
>
> Sections with this band run programs from a terminal. You can skip them and still follow every concept.

Which version this describes

The cover names the commit of the `main` branch this edition describes, and the date it was built. The latest edition is always at github.com/glitchwizard/saulitwerx, in `docs/manual/`. **What's new**, near the back, lists what each edition added.

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PART I

Getting started

What Saulitwerx is, what it can do today, how to install it, and a first session that ends with a picture of a real part.

CHAPTER 1

Welcome to Saulitwerx

1.1 What Saulitwerx is

Saulitwerx is a program for designing solid mechanical parts on a computer: brackets, plates, shafts, housings, and anything else you might machine, cast or 3D-print. Programs like this are called *CAD* (computer-aided design). Saulitwerx is free and open source, so anyone can use it, study it, change it, and build it into other software.

Saulitwerx is a *parametric* modeler. You don't sculpt a shape by pushing points around. Instead you describe how to build it, step by step, with measurements: "draw a 40 by 25 mm rectangle, push it up 5 mm, cut a 10 mm hole through the middle." Saulitwerx remembers those steps and their numbers. Change the hole to 12 mm later, and the part rebuilds itself to match. Chapter 4 explains how that works.

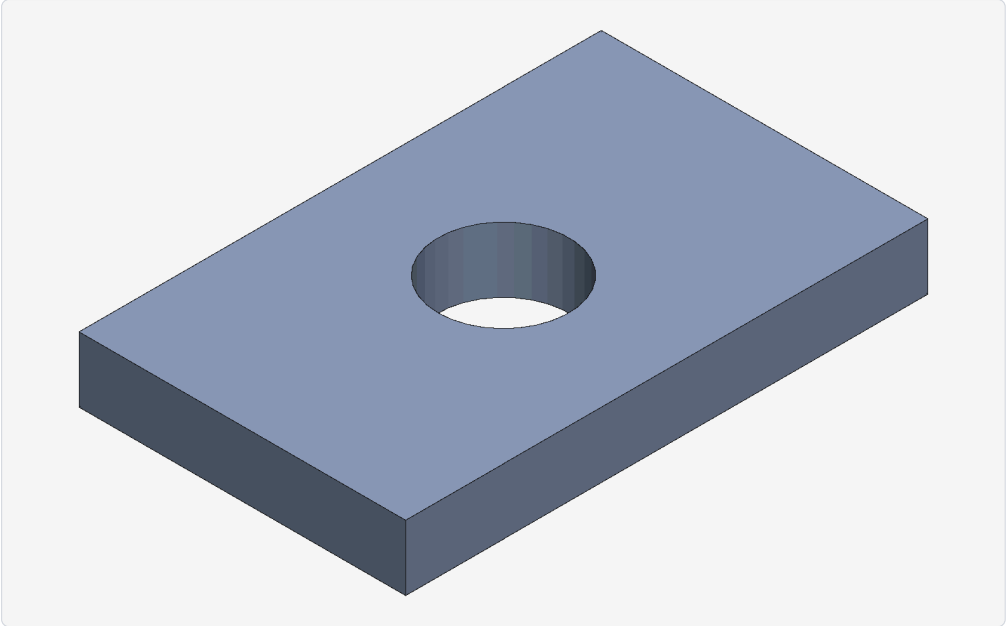


Figure 1.1 A plate with a hole, made by Saulitwerx's geometry engine and drawn by its renderer. You will make this part yourself in Chapter 3.

1.2 What it can do today

Saulitwerx is built in small, tested steps, in the order of a public plan (the *roadmap*). Here is where it stands for this edition.

Area	What exists	Use it
Solids	Make boxes, cylinders and spheres; combine them (add, subtract, keep the overlap); measure volume, surface area and size.	NOW

Area	What exists	Use it
Exchange	Save a solid as a STEP file, and read STEP files back.	NOW
Pictures	Draw a solid or a STEP file to a PNG picture with `swx-qa render`, and a sketch with `swx-qa sketch`.	NOW
Sketching	Lines, circles, arcs and points, with thirteen kinds of constraint (from horizontal and coincident to tangent and symmetric), a solver that moves the shape to satisfy them, a check of how fully a sketch is defined, and the regions a sketch encloses, ready to become solids.	NOW
History	Keep features in order with named parameters, and work out which features a change affects.	NOW
Extrude	Push a flat outline of lines, arcs and circles, with holes, into a solid, on any plane, one way or both, or cut right through a part.	NOW
Revolve	Turn a flat outline around an axis into a round solid, by a full turn or part of one.	NOW
Desktop app	Windows, menus, a 3D view and mouse control.	LATER
Part files	Saving your design with its full history in a Saulitwerx part file.	LATER
Features	Extrude and revolve driven by sketches and recorded in the history; fillet, chamfer, hole, shell and more.	LATER
Assemblies, drawings	Putting parts together, and making 2D shop drawings.	LATER

Everything marked NOW is a building block that programmers can call today. The **Tasks** part shows each one with a real command or a short program you can run. Everything marked LATER is described in **Coming next** at the back, and nowhere else.

1.3 Who should use it now

- **If you are new to CAD,** read Parts I and II. You will learn the ideas every parametric modeler is built on, and you will be ready the day the desktop app arrives.
- **If you are comfortable in a terminal,** install Saulitwerx (Chapter 2) and follow Chapter 3. It takes about 15 minutes and ends with a picture of a part you built.
- **If you write software,** the **For developers** sections show how to call each building block from Rust.

> **NOTE**
>
> Saulitwerx reads and writes only open, documented file formats. It will never open another program's own native files. To bring in a design from elsewhere, export it as STEP first.

CHAPTER 2

Installing Saulitwerx

There is no installer yet. Until the first release, you build Saulitwerx from its source code. That sounds harder than it is: three tools to install, then one command builds everything. Plan on about 10 minutes, most of it waiting for downloads.

YOU WILL	Install the tools Saulitwerx needs, download its source code, build it, and check that it works.
YOU NEED	A computer running Linux (these steps were tested on Ubuntu 24.04), an internet connection, and permission to install software (`sudo`).
TIME	About 10 minutes.

2.1 What gets installed

Tool	Why Saulitwerx needs it	Required?
Git	Downloads the source code and keeps it up to date.	Yes
Rust (1.80 or newer)	The programming language Saulitwerx is written in. Its tool `cargo` builds and runs the program.	Yes
C++ compiler	Builds a small bridge between Saulitwerx and the geometry engine.	For solids and STEP
OpenCascade (OCCT) 7.6 or newer	The open-source geometry engine that does the solid modeling.	For solids and STEP

Without OpenCascade you can still sketch, try the document model, and draw the built-in sample parts. To make your own solids or read and write STEP files, you need it.

2.2 Install the tools

1. Open a terminal. On Ubuntu, press `Ctrl` + `Alt` + `T`. A window with a command prompt appears.
2. Install Git, a C++ compiler and OpenCascade:

   TERMINAL

   ```
   sudo apt-get update
   sudo apt-get install -y git build-essential curl \
      libocct-foundation-dev libocct-modeling-data-dev \
      libocct-modeling-algorithms-dev libocct-visualization-dev \
      libocct-data-exchange-dev
   ```

   Type your password when asked. The terminal prints a list of packages and ends at the prompt again.
3. Install Rust with its official installer, and accept the default choices when it asks:

   TERMINAL

   ```
   curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
   ```

It ends with `Rust is installed now. Great!`

4. Close the terminal and open a new one, so it finds the new tools. Then check them:

```
TERMINAL
cargo --version
```

You should see a version line such as `cargo 1.97.0 (…)`. Any version 1.80 or newer works.

2.3 Get the source and build it

1. Download the source code into a folder named `saulitwerx`, and go into it:

```
TERMINAL
git clone https://github.com/glitchwizard/saulitwerx
cd saulitwerx
```

2. Build Saulitwerx and run its own checks:

```
TERMINAL
cargo test --workspace
```

The first build downloads a few small libraries and takes a minute or two. It ends with several lines that start `test result: ok.` and none that say `FAILED`.

3. Build the picture tool with OpenCascade support:

```
TERMINAL
cargo build --release -p swx-qa --features occt
```

It ends with a line that starts `Finished`. The tool is now at `target/release/swx-qa`.

4. Check that the picture tool runs:

```
TERMINAL
target/release/swx-qa --help
```

You should see its help text, which starts `Usage: swx-qa render`.

RESULT Saulitwerx is built, its tests pass, and the `swx-qa` picture tool is ready. Go on to Chapter 3.

> **TIP** **Staying up to date**
>
> Saulitwerx changes most nights. To get the newest version, go into the `saulitwerx` folder and run `git pull`, then build again.

> **CAUTION** **Commands run from the `saulitwerx` folder**
>
> Every command in this manual assumes your terminal is in the `saulitwerx` folder. If a command says it can't find a file, run `cd saulitwerx` first. Chapter 12 lists other problems and their fixes.

2.4 Other systems

The build is regularly checked only on Ubuntu Linux. Saulitwerx aims to run on macOS and Windows too, and the same steps should work there once Rust and OpenCascade are installed, but they are untested. Without OpenCascade, leave out `--features occt` in step 3.

CHAPTER 3

Your first part

In this tutorial you draw a sample part, then build a real one, a plate with a hole through it, measure it, save it in a standard file format, and draw it. Along the way you will meet the ideas the rest of the manual builds on.

YOU WILL	Draw a built-in sample part, build a plate with a hole, check its measurements, save it as a STEP file, and draw it.
YOU NEED	Saulitwerx built with OpenCascade support (Chapter 2), and a terminal in the `saulitwerx` folder.
TIME	About 15 minutes.

3.1 Draw a sample part

Saulitwerx comes with three sample parts built in: a cube, a cylinder and an L-shaped bracket. They need nothing else, so they are a quick way to check that drawing works.

1. Draw the cube to a picture file named `cube.png`:

   TERMINAL

   ```
   target/release/swx-qa render --fixture cube -o cube.png
   ```

   The tool answers with one line:

   OUTPUT

   ```
   wrote cube.png (800x600, 27669 bytes)
   ```

2. Open `cube.png` in any image viewer. On Ubuntu, type `xdg-open cube.png`. You should see a grey-blue cube on a light grey background, like Figure 3.1.

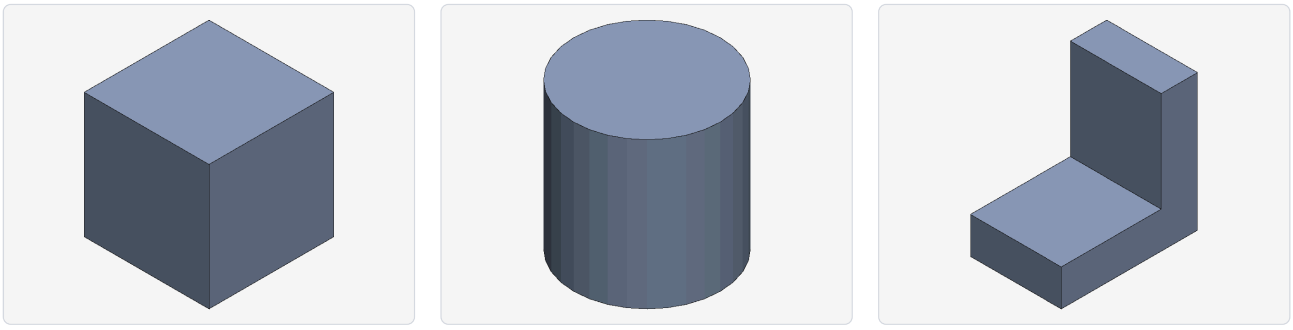


Figure 3.1 The three built-in sample parts: `cube`, `cylinder` and `l-bracket`.

The picture is always drawn from the same corner, looking down at an angle, so that three sides of a box show at once. Each flat face gets one even shade, and sharp edges get a dark outline. This kind of view is called *isometric*.

> **TIP**
>
> Try `--fixture cylinder` and `--fixture l-bracket` too. Add `--size 1600x1200` for a bigger, sharper picture.

3.2 Build a plate with a hole

Now build a real part. Saulitwerx doesn't have a window to draw in yet, so this part comes from a small program that ships with Saulitwerx. The program asks the geometry engine for three things:

1. a *solid* block 40 mm long, 25 mm wide and 5 mm thick;
2. a cylinder 10 mm across, standing where the hole should go, a little taller than the plate;
3. the block **minus** the cylinder. Taking one solid away from another is called a *cut*.

1. Run the program, and tell it to save the part as `plate.step`:

   TERMINAL

   ```
   cargo run -p swx-kernel --features occt --example plate_with_hole -- plate.step
   ```

   After a few seconds of building, it prints the part's measurements. The geometry engine also prints a short report about writing the file, framed in rows of `*`; you can ignore it.

   OUTPUT

   ```
   volume        4607.30 mm³
   surface area  2650.00 mm²
   faces         7
   bounding box  [0.0, 0.0, 0.0] to [40.0, 25.0, 5.0]
   …  (the geometry engine's file report)
   wrote plate.step (read back: 4607.30 mm³, 7 faces)
   ```

2. Check the numbers. The block holds $40 \times 25 \times 5 = 5000$ mm³. The hole removes $\pi \times 5^2 \times 5 \approx 392.7$ mm³. $5000 - 392.7 = 4607.3$, which is what the program reports. The part has 7 faces: top, bottom, four sides and the inside of the hole.

3. Draw the part:

   TERMINAL

   ```
   target/release/swx-qa render --step plate.step --size 1600x1000 -o plate.png
   ```

   The tool answers `wrote plate.png (1600x1000, …)`.

4. Open `plate.png`. You should see the plate of Figure 3.2, with the hole showing light background through it.

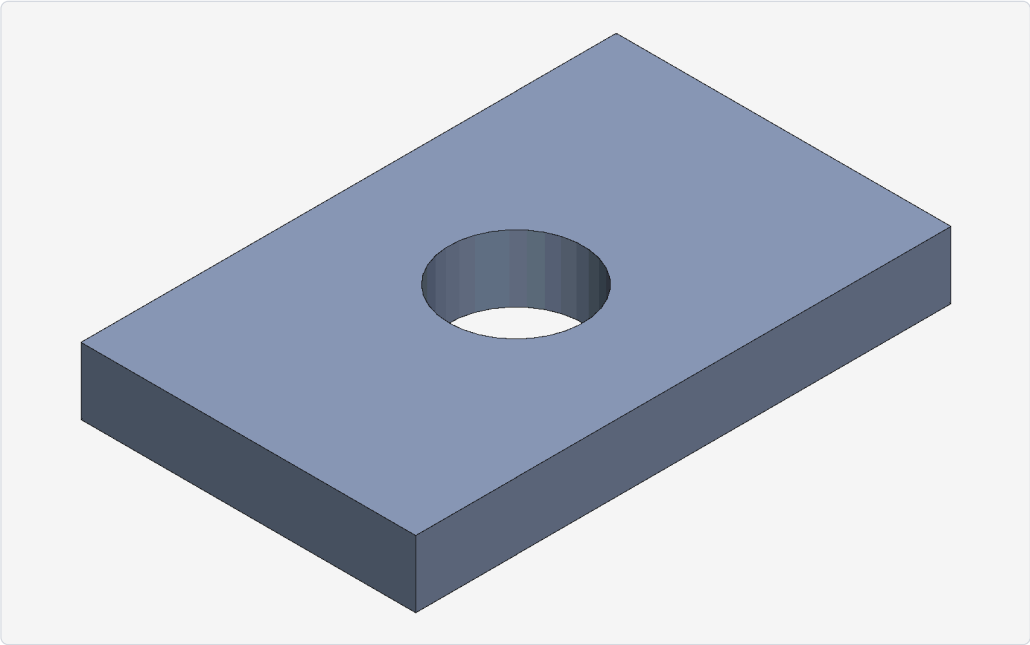


Figure 3.2 The plate you built, drawn from the STEP file the program saved.

RESULT You built a solid part, checked its volume, saved it as `plate.step`, and drew it as `plate.png`.

3.3 What you just did

The file `plate.step` is in the *STEP* format, an international standard (ISO 10303) for exchanging 3D parts. Almost every CAD program can open it, so you can take this plate into another tool, send it to a machine shop, or prepare it for 3D printing.

What you did **not** see yet is the parametric part of Saulitwerx: the numbers 40, 25, 5 and 10 are typed into the program, not stored as named, editable parameters. That comes with the part features and part files on the roadmap. Chapter 4 explains the ideas behind it, and Chapter 10 shows the piece of it that already works.

> **FOR DEVELOPERS**
>
> The program you ran is `crates/swx-kernel/examples/plate_with_hole.rs`. Change `5.0` (the hole radius) in the `make_cylinder` line to `8.0`, run step 1 again, and the volume drops to 3994.69 mm³. Chapter 6 explains each call.

PART II

Concepts

The ideas behind parametric CAD, in plain words. Read this before the task chapters; every procedure later relies on these terms.

CHAPTER 4

How parametric CAD works

Every parametric modeler, Saulitwerx included, is built on a handful of ideas. Once you know them, the commands make sense on their own. This chapter explains each idea in the order you meet it when you design a part. For each one it says what it is, why it matters, and where it stands in Saulitwerx today.

4.1 Solids

A *solid* is a closed 3D shape with an inside and an outside, like a real object. Because Saulitwerx knows what is inside, it can tell you how much material a part holds (its *volume*), how much surface it has to paint (its *surface area*), and how big a box it would fit in (its *bounding box*).

The outside of a solid is made of *faces*: flat or curved patches of surface. Where two faces meet there is an *edge*, and where edges meet there is a corner point, a *vertex* (plural: vertices). The plate from Chapter 3 has 7 faces: top, bottom, four sides, and the round inside of the hole.

Faces, edges and vertices are what you will later pick to say “round this edge” or “sketch on this face”, so it helps to see how they hang together. Each face is bounded by edges, and each edge ends at vertices. A box has 6 faces, 12 edges and 8 vertices; every edge sits between exactly two faces.

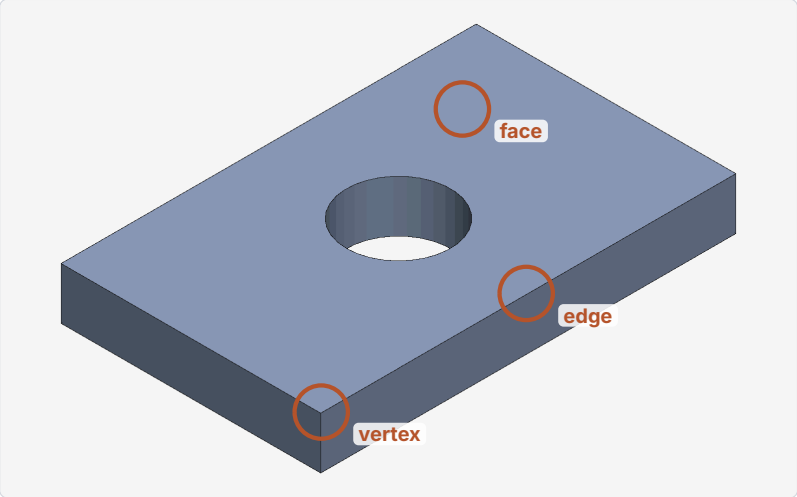


Figure 4.1 A solid, with one face, one edge and one vertex ringed. Faces are shaded and edges outlined. Volume 4607.30 mm³, 7 faces.

In Saulitwerx today: the geometry engine makes boxes, cylinders and spheres, and measures volume, surface area, bounding box and face count. It can also count a solid’s faces, edges and vertices, and list which ones touch (Section 6.3). Chapter 6 shows how.

4.2 Units

Every length in a part has a *unit*. In Saulitwerx the default is the millimeter (mm), and all of today's building blocks work in millimeters: a box 40 long is 40 mm long, and volumes come out in mm³. Angles are in degrees where you type them, and in radians inside the sketch geometry.

In Saulitwerx today: everything is in millimeters. STEP files are written in millimeters too. Choosing units per part comes with part files.

4.3 Sketches

Most solids start as a flat drawing. A *sketch* is a 2D drawing on a flat plane, made of simple pieces: points, straight lines, circles and arcs. Together these are called sketch *geometry* or *entities*.

A sketch for the plate would be a rectangle (four lines) and a circle. You don't need to draw it accurately. A rough shape is fine, because the next two ideas, constraints and dimensions, make it exact.

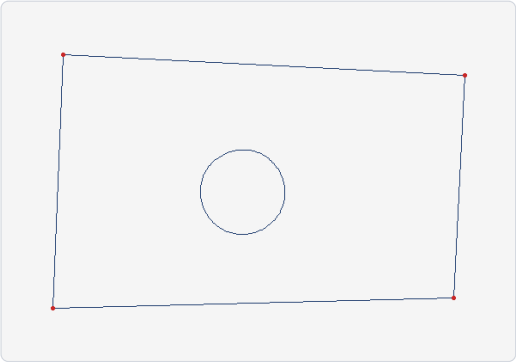


Figure 4.2 A rough sketch of the plate: four lines and a circle, drawn by eye. The red dots are the ends of the lines.

In Saulitwerx today: sketches with points, lines, circles and arcs exist as a building block (Chapter 5). There is no drawing window yet, so sketches are written as a short program, and `swx-qa sketch` draws them to a picture (Section 9.5).

4.4 Constraints

A *constraint* is a rule about the shape of a sketch: "this line is horizontal," "these two corners touch," "this point stays at the origin." Constraints carry the **intent** of a design. If you later change a size, a line that must stay horizontal stays horizontal.

Constraint	Rule it keeps
Coincident	Two points are at the same place. Use it to join the end of one line to the start of the next.
Horizontal	A line runs left–right; or two points are at the same height.
Vertical	A line runs up–down; or two points are directly above each other.
Fixed	A point stays at exact coordinates. Use it to anchor a sketch.
Parallel	Two lines point the same way, like opposite sides of a rectangle.
Perpendicular	Two lines meet at a right angle (90°), like the corner of a rectangle.
Equal	Two lines are the same length, or two circles or arcs are the same size.
Midpoint	A point sits exactly halfway along a line.
Symmetric	Two points are mirror images of each other across a line.
Tangent	A line just touches a circle or arc without cutting into it; or two circles or arcs just touch.

Some constraints are about direction (horizontal, vertical, parallel, perpendicular), some about position (coincident, fixed, midpoint, symmetric), and some about how curves meet (tangent) or match (equal). Parallel and perpendicular are relative: a parallel pair can lean at any angle, as long as both lean the same way. That is why a rectangle usually gets one horizontal side, to say which way is level, and perpendicular and parallel for the rest.

You don't have to add every constraint by hand. Saulitwerx can add the ones your drawing already almost shows, such as a corner that nearly touches or a line that is nearly level (Section 5.8).

Saulitwerx satisfies constraints with a *solver*: a calculation that nudges every point until all the rules hold at once. You give it a rough drawing and the rules; it gives back the exact shape.

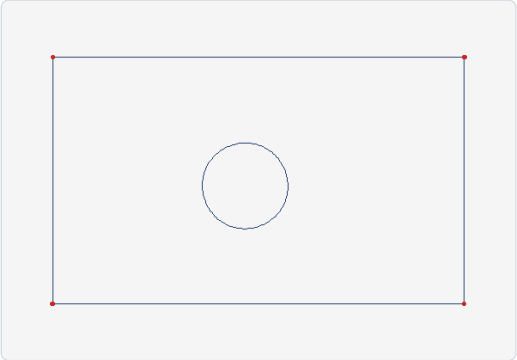


Figure 4.3 The same sketch after the solver applied coincident, horizontal, vertical and fixed. It is now a true rectangle, but its size is still whatever the rough drawing happened to be.

4.5 Dimensions

A *dimension* is a constraint with a number: a length, a distance, a radius or a diameter. "The bottom line is 40 mm long." "The hole is 10 mm across." Dimensions fix the size of a sketch, the way the plain constraints above fix its shape.

Dimension	Number it sets
Distance	The straight-line distance between two points, for example the two ends of a line.
Radius	The radius of a circle or arc.
Diameter	The diameter of a circle or arc: twice the radius.

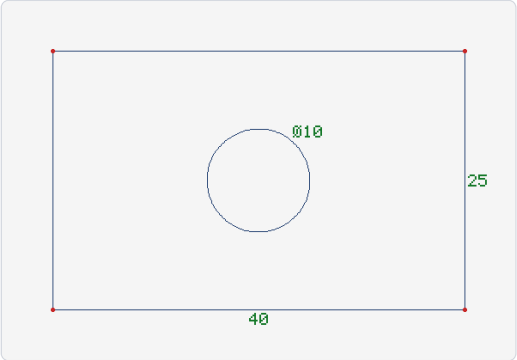


Figure 4.4 With dimensions added, the plate sketch is exactly 40 × 25 mm with a ⌀10 mm hole. Each green number is a dimension.

4.6 Degrees of freedom

Each way a piece of a sketch can still move is called a *degree of freedom*, or DOF for short. A free point on a sheet has two (left–right and up–down). A free line has four (two for each end); a free circle has three (its center's two, and its size). Each constraint takes some away: fixed takes two from a point, horizontal takes one from a line.

Counting what is left tells you how finished a sketch is. A sketch is in one of four states: *under-constrained*, *fully constrained*, *over-constrained* or *conflicting*.

State	What it means
Under-constrained	Something can still move. Add constraints or dimensions until the count reaches zero.
Fully constrained	Nothing can move, and no rule repeats another. This is the goal.
Over-constrained	At least one rule repeats what other rules already say, but they all agree. The repeated rule is *redundant*: harmless today, but it will fight you as soon as you change one of the sizes it repeats.
Conflicting	Redundant rules that disagree, such as two sides of a rectangle dimensioned 40 and 45 mm. No shape can satisfy them, so the solver stops short.

The plate sketch starts with 19 degrees of freedom: four lines and a circle. Figure 4.3 still has 5 (the rectangle's width and height, and the circle's position and size); Figure 4.4 has none. Section 5.3 shows how to get these numbers.

> **TIP**
>
> Constrain the shape first (coincident, horizontal, vertical), then add dimensions. Then anchor one point with fixed, so the whole sketch can't slide around.

In Saulitwerx today: all of the constraints and dimensions in these tables work, and the solver reports whether it found a shape that meets them. A degree-of-freedom report gives the count and the state, and names each redundant constraint (Section 5.3). Constraints can name every point of a sketch: a point, a line's ends, a circle's or arc's center, and an arc's two ends, so lines and arcs join into closed outlines such as a slot (Section 5.6).

4.7 Profiles and regions

A sketch on its own is just lines. To become a solid, it has to enclose an area that a feature can push into 3D. A closed chain of sketch pieces, where each one ends exactly where the next begins, is a *loop*. The flat area a loop encloses is a *region*, and the regions of a sketch together are its *profile*.

A region can have holes. In the plate sketch the rectangle is one loop and the circle is another, inside it. Saulitwerx works out that the circle is a hole, so the plate's region is the rectangle **minus** the circle: 1000 − 78.54 = 921.46 mm². Pushed up 5 mm, that region is exactly the plate of Chapter 3, hole included, with no separate cut.

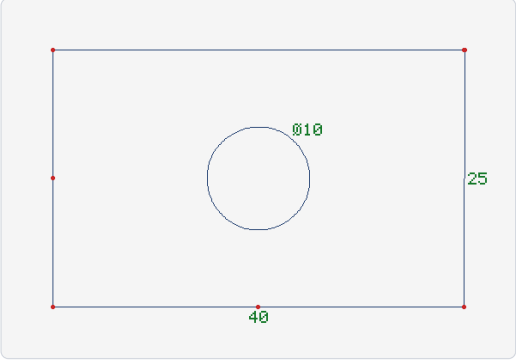


Figure 4.5 Two loops, one region: the rectangle is its outside edge and the circle its hole.

Pieces that don't close a loop, like a line with a free end, are *open edges*. They are left out of every region and listed separately, so you can find the gap. Pieces that cross each other are an error: Saulitwerx won't guess which of the shapes they form you meant.

In Saulitwerx today: profiles are worked out from any sketch of lines, circles and arcs, with holes, nested islands and open edges (Section 5.7). The geometry engine can extrude an outline with holes into a solid (Chapter 7), or revolve one around an axis (Chapter 8), but the outline is typed in by hand: a sketch's regions can't be passed to either yet.

4.8 Parameters

A *parameter* is a named number, like `thickness = 5` or `hole_dia = 10`. Features and dimensions refer to parameters by name instead of holding a bare number. Change the parameter, and everything that uses it follows.

This is the "parametric" in parametric CAD. One plate design, with its thickness as a parameter, is really a whole family of plates: set `thickness` to 3, 5 or 8 and rebuild.

Parameters come in two kinds. Each dimension owns a *model parameter*, named `d0`, `d1`, … in the order the dimensions were made. You add *user parameters* yourself, with names you choose. Rename `d0` to `thickness`, and everything that uses it follows.

An *expression* is a small formula with units, such as `d1 / 2 + 5 mm`: half of `d1`, plus 5 millimeters. Saulitwerx checks that the units make sense, so adding a length to an angle is an error, not a silent wrong answer.

In Saulitwerx today: parts keep a parameter table with both kinds, units and comments, and features refer to parameters by name (Section 10.3). A sketch's distance, radius and diameter dimensions can each make their own model parameter and follow it when it changes (Section 10.4). A parameter can hold an expression, and follows the parameters it names (Section 10.6).

4.9 Features

A *feature* is one step in building a part: a sketch, or something done to the solid, such as pushing a sketch out into 3D or cutting a hole. Each feature records **what** it does and **which** sketch and parameters it uses, not just the shape that resulted.

Figure 4.6 shows the idea of a cut, done directly with the geometry engine: a block, minus a cylinder, gives the plate. An **extrude** can make the same plate in one step, from the outline of the rectangle and its hole (Chapter 7). Recording those steps as features in a part's history comes later.

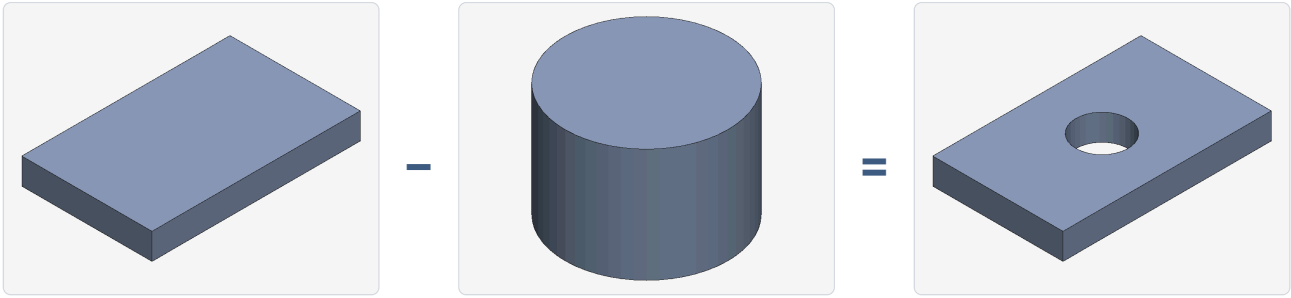


Figure 4.6 A cut takes one solid away from another. Each picture is scaled to fill its frame, so the cylinder looks larger than it is.

There are three ways to combine two solids. Together they are called *Boolean operations*:

- **Union** (join): keep everything that is in either solid.
- **Cut** (subtract): keep the first solid, minus anything the second one overlaps.
- **Intersect**: keep only the part that is in both.

4.10 History

A part's *history*, also called its *feature tree*, is its list of features in the order they were made. Each feature can use only what came before it: a cut needs a solid to cut into, and an extrude needs a sketch to push.

#	Feature	Uses
0	Sketch1	nothing: a rectangle on a plane
1	Extrude1	Sketch1, and the parameter `thickness`
2	Sketch2	nothing: a profile for a pin
3	Revolve1	Sketch2, and the parameter `sweep`

Because the history remembers how each feature was made, Saulitwerx can **rebuild** a part after a change. It doesn't rebuild everything. It works out which features depend on what changed, and redoes only those. Change `thickness`, and only Extrude1 rebuilds; edit Sketch2, and Sketch2 and Revolve1 rebuild.

In Saulitwerx today: the history and the "what needs rebuilding" logic exist (Chapter 10), using the exact example in the table above. Running the rebuild on real geometry comes with the part features.

4.11 Putting it together

Here is how the ideas line up when you design a part, once the app exists:

1. Make a **sketch** on a plane, roughly the shape you want.
2. Add **constraints** to fix its shape, and **dimensions** to fix its size, until no degrees of freedom are left.
3. Turn the sketch into a **solid** with a **feature**, such as extrude.
4. Add more features (cuts, holes, rounded edges) to finish the part. They form its **history**.
5. Change a **parameter** or a dimension later, and the part rebuilds to match.

PART III

Tasks

How to do things, one task per section. Until the desktop app arrives, every task here runs a real command or a small program, and is marked for developers.

CHAPTER 5

Sketching

> **FOR DEVELOPERS**
>
> Until the desktop app's sketch mode arrives, sketches are written as a short Rust program against the `swx-sketch` library. This chapter runs one that ships with Saulitwerx.

5.1 Solve a sketch

The sketch for the plate of Chapter 3 is a 40 × 25 mm rectangle with a 10 mm circle in the middle. In this task you run a program that draws that sketch **roughly**, adds the constraints and dimensions, and asks the solver for the exact shape.

YOU WILL	Run the sample sketch, read the solver's report, and see the rough shape become exact.
YOU NEED	Saulitwerx built (Chapter 2). OpenCascade is not needed.
TIME	5 minutes.

1. Run the sample sketch:

   TERMINAL
   ```
   cargo run -p swx-sketch --example plate_sketch
   ```

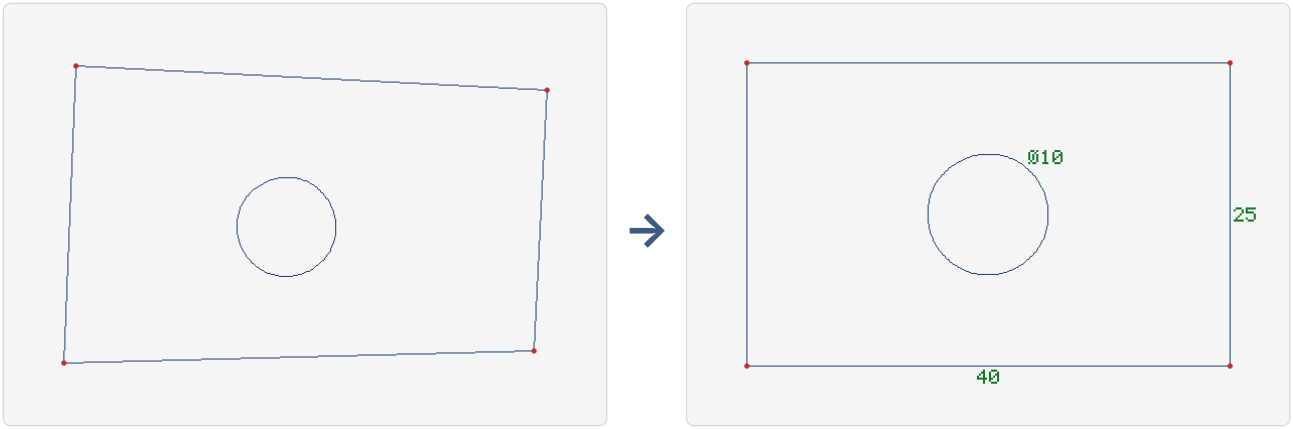
   It prints the solver's report, then each line and the circle:

   OUTPUT
   ```
   solved: true (Converged, 3 iterations, residual 5.3e-14)
   bottom (  0.00,   0.00) -> ( 40.00,   0.00)  length 40.00 mm
   right  ( 40.00,   0.00) -> ( 40.00,  25.00)  length 25.00 mm
   top    ( 40.00,  25.00) -> (  0.00,  25.00)  length 40.00 mm
   left   (  0.00,  25.00) -> (  0.00,   0.00)  length 25.00 mm
   hole   center (20.00, 12.50)  diameter 10.00 mm
   ```

2. Read the first line. `solved: true` and `Converged` mean the solver found a shape that meets every rule. `residual` is how far off it still is, here about 0.00000000000005 mm: exact for any practical purpose.

3. Compare the result with the rough drawing in the program (below). The bottom line was drawn 38 mm long and tilted; it is now exactly 40 mm and level.

RESULT A fully sized, square-cornered sketch: a 40 × 25 mm rectangle with its lower-left corner at the origin, and a 10 mm circle centered at (20, 12.5).



Before: the rough drawing After: solved

Figure 5.1 The sample sketch before and after solving. Section 9.5 shows how to draw pictures like these.

5.2 How the sample sketch is built

The program is `crates/swx-sketch/examples/plate_sketch.rs`. It works in four stages, the same ones you will use in the app. The code below is shortened from the file.

1. Draw roughly. Four lines and a circle, deliberately off-size:

```
let bottom = s.add(Geometry::Line(Line::new(p(0.0, 0.0)?, p(38.0, 1.0)?)?))?;
let right = s.add(Geometry::Line(Line::new(p(38.0, 1.0)?, p(39.0, 22.0)?)?))?;
let top = s.add(Geometry::Line(Line::new(p(39.0, 22.0)?, p(1.0, 24.0)?)?))?;
let left = s.add(Geometry::Line(Line::new(p(1.0, 24.0)?, p(0.0, 0.0)?)?))?;
let hole = s.add(Geometry::Circle(Circle::new(p(18.0, 11.0)?, 4.0)?))?;
```

2. Fix the shape. Join each line's end to the next line's start with **coincident**, then make two lines horizontal and two vertical:

```
for (a, b) in [(bottom, right), (right, top), (top, left), (left, bottom)] {
    s.add_constraint(Constraint::Coincident(PointRef::End(a), PointRef::Start(b)))?;
}
s.add_constraint(Constraint::Horizontal(bottom))?;
s.add_constraint(Constraint::Vertical(left))?;   // and likewise top, right
```

3. Anchor and size it. Pin the lower-left corner to the origin with **fixed**, and set two **distances** and the hole's **diameter**:

```
s.add_constraint(Constraint::Fixed { point: PointRef::Start(bottom), x: 0.0, y: 0.0 })?;
s.add_constraint(Constraint::Distance {
    a: PointRef::Start(bottom), b: PointRef::End(bottom), value: 40.0 })?;
s.add_constraint(Constraint::Diameter { geom: hole, value: 10.0 })?;
```

4. Solve. One call moves everything into place and returns a report:

```
let report = s.solve(SolveOptions::default())?;
```

A `PointRef` names a point the constraint acts on: `Start(line)`, `End(line)`, `Center(circle)`, `ArcStart(arc)`, `ArcEnd(arc)` or `Point(point)`. Section 11.6 lists every entity and constraint with its exact form.

5.3 Check how fully a sketch is defined

A sketch that solves is not necessarily finished: parts of it may still be free to move, or some rules may repeat each other. A degree-of-freedom report (Section 4.6) tells you which. In this task you run a program that builds the plate sketch one group of rules at a time and checks it after each.

YOU WILL	Watch the degrees of freedom of a sketch fall to zero, then see what a repeated or contradicting dimension looks like.
YOU NEED	Saulitwerx built (Chapter 2). OpenCascade is not needed.
TIME	5 minutes.

1. Run the sample:

   TERMINAL

   ```
   cargo run -p swx-sketch --example sketch_dof
   ```

   It prints one line per stage, with the number of equations the rules make, the degrees of freedom left, and the state:

   OUTPUT

   ```
   1 rough drawing            equations  0  dof 19  UnderConstrained
   2 + coincident x4          equations  8  dof 11  UnderConstrained
   3 + horizontal, vertical   equations 12  dof  7  UnderConstrained
   4 + fixed corner           equations 14  dof  5  UnderConstrained
   5 + two distances          equations 16  dof  3  UnderConstrained
     solve: Converged
   6 + hole, then solve       equations 19  dof  0  FullyConstrained
     solve: Converged
   7 + distance top 40        equations 20  dof  0  OverConstrained
                              redundant: distance top 40
     solve: Stagnated
   7 + distance top 45        equations 20  dof  0  Conflicting
                              redundant: distance top 45
   ```

2. Read stages 1 to 6. The four lines and the circle start with 19 degrees of freedom. Each group of rules takes some away, and after the hole is placed and sized none are left: `FullyConstrained`.
3. Read the first stage 7. A third distance, 40 mm on the top line, repeats what the bottom line and the horizontal and vertical rules already say. The solver still succeeds, but the report says `OverConstrained` and names the repeated rule.
4. Read the second stage 7. With 45 mm on the top line instead, no rectangle can meet every rule. The solver stops with `Stagnated`, and the report says `Conflicting` and names the 45 mm distance.

RESULT You can tell an unfinished sketch (`UnderConstrained`, with the number of degrees of freedom still open) from a finished one (`FullyConstrained`), and spot a repeated or contradicting rule.

How the check works

The program is `crates/swx-sketch/examples/sketch_dof.rs`. One call on the sketch makes the report:

```rust
use swx_sketch::dof::DofOptions;

let r = s.dof(DofOptions::default())?;
println!("dof {} {:?}", r.report.dof, r.report.state);
for id in &r.redundant_constraints {
    println!("redundant: {}", names[id.0 as usize]);
}
```

`r.report` holds `dof` and the `state`. `r.redundant_constraints` lists the repeated rules as constraint ids: the number `add_constraint` gave each one, counting from 0 in the order you added them. The sample keeps a list of names in the same order, so `names[id.0]` prints the rule in words. A rule that adds two equations, such as a repeated coincident, is listed once. Section 11.9 lists every field and setting.

> **TIP** **Solve first, then check**
>
> The difference between over-constrained and conflicting is whether the rules are met. Run the check after `solve`: on a rough drawing that hasn't been solved yet, a harmless repeated rule can show as `Conflicting`.

> **CAUTION** **Only the later rule is named**
>
> When two rules repeat each other, the report names the one you added **later**, because the earlier one was still independent when it arrived. The fix may be to delete either one.

5.4 Change a dimension

1. Open `crates/swx-sketch/examples/plate_sketch.rs` in a text editor.
2. Find `value: 40.0` and change it to `value: 60.0`. Save the file.
3. Run the sample again (step 1 above). The bottom and top lines now report `length 60.00 mm`; the right line moves to x = 60. The hole stays at (20, 12.5), because it is fixed there, not centered.
4. Change the value back to `40.0` when you are done.

> **TIP**
>
> The hole stayed put because it is fixed at (20, 12.5). To keep it centered whatever the width, use midpoint instead (Section 5.5).

5.5 Keep a hole centered

A good sketch says what you **mean**. If the hole belongs in the middle of the plate, the sketch should say "in the middle", not "at x = 20". Then a wider plate keeps its hole centered with no extra work. The tool for this is the *midpoint* constraint, which keeps a point exactly halfway along a line.

YOU WILL	Center a hole with midpoint constraints, then change the plate's width and watch the hole follow.
YOU NEED	Saulitwerx built (Chapter 2). OpenCascade is not needed.
TIME	5 minutes.

1. Run the centered-plate sample:

   TERMINAL

   ```
   cargo run -p swx-sketch --example centered_plate
   ```

   It solves the sketch, checks its degrees of freedom, and prints the sizes and the region an extrude would use:

OUTPUT

```
solved: true (Converged), dof 0 FullyConstrained
bottom length 40.00 mm
right  length 25.00 mm
hole   center (20.00, 12.50)  diameter 10.00 mm
regions: 1
  region 0: outer edges 4, holes 1, area 921.46 mm²
open edges: 0
```

2. Check the first line. `FullyConstrained` with `dof 0` means every piece is pinned down, including the hole, even though no rule gives the hole's coordinates.

3. Run it again for a plate 60 mm wide. The two numbers after `--` are the width and the height:

TERMINAL

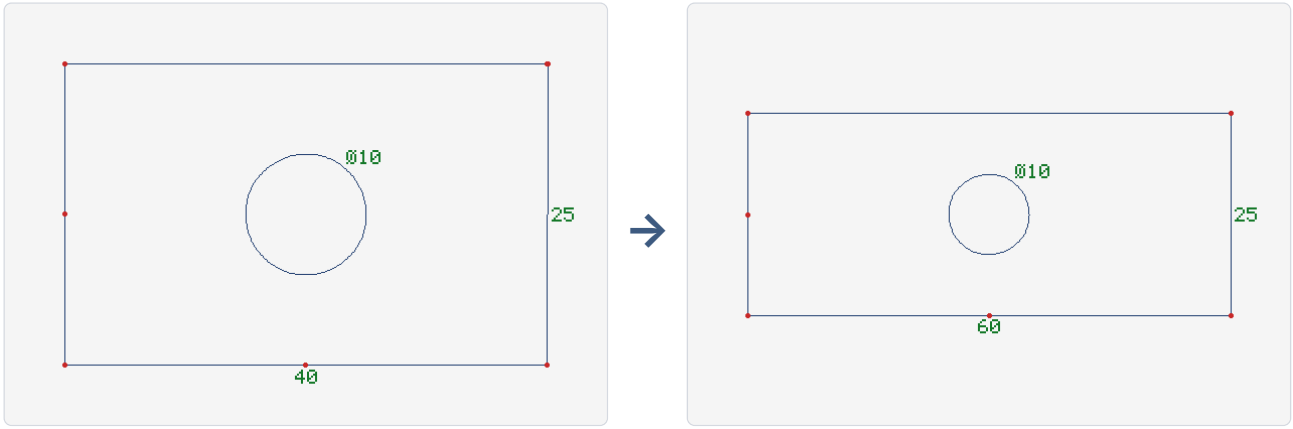
```
cargo run -p swx-sketch --example centered_plate -- 60 25
```

OUTPUT

```
solved: true (Converged), dof 0 FullyConstrained
bottom length 60.00 mm
right  length 25.00 mm
hole   center (30.00, 12.50)  diameter 10.00 mm
regions: 1
  region 0: outer edges 4, holes 1, area 1421.46 mm²
open edges: 0
```

The hole has moved to x = 30, the middle of the new width.

RESULT A plate whose hole stays in the middle whatever its width or height, and a sketch that is fully constrained.



40 mm wide 60 mm wide

Figure 5.2 The same sketch with only the width changed. The red dots at the middle of the bottom and left sides are the helper points that center the hole.

How the centered plate is built

The program is `crates/swx-sketch/examples/centered_plate.rs`. It starts from the same rough drawing as the plate sketch, and changes two things.

Square corners without four direction rules. Only the bottom is horizontal. The left side is *perpendicular* to it, and each side is *parallel* to the one opposite:

```
s.add_constraint(Constraint::Horizontal(bottom))?;
s.add_constraint(Constraint::Perpendicular(bottom, left))?;
```

```
s.add_constraint(Constraint::Parallel(top, bottom))?;
s.add_constraint(Constraint::Parallel(right, left))?;
```

Center the hole with two helper points. A point entity sits at the middle of the bottom side, and another at the middle of the left side. The hole's center is lined up straight above the first and level with the second:

```
let mid_bottom = s.add(Geometry::Point(p(19.0, 0.0)?))?;
let mid_left = s.add(Geometry::Point(p(0.0, 12.0)?))?;
s.add_constraint(Constraint::Midpoint { point: PointRef::Point(mid_bottom), line: bottom })?;
s.add_constraint(Constraint::Midpoint { point: PointRef::Point(mid_left), line: left })?;
s.add_constraint(Constraint::VerticalPoints(
    PointRef::Point(mid_bottom), PointRef::Center(hole)))?;
s.add_constraint(Constraint::HorizontalPoints(
    PointRef::Point(mid_left), PointRef::Center(hole)))?;
```

Helper points are safe to add: they are not part of any outline, so the profile ignores them.

> **TIP** **Why not a diagonal?**
>
> Drawing the rectangle's diagonal and putting the hole's center at its midpoint would also work for the solver. But the diagonal would cut through the hole and the plate, and profiles can't be found from pieces that cross (Section 5.7). Helper points avoid that.

The *symmetric* constraint is the other way to say "centered": `Symmetric { a, b, line }` makes points `a` and `b` mirror images across `line`. It suits pairs, such as two holes placed evenly either side of a center line.

5.6 Draw a slot

A slot is a rounded-end opening, like the hole a bolt slides along. Its outline is two straight sides and two half circles, and the sides must run smoothly into the curves, with no corner. Three constraints say that. Coincident joins each side's ends to the arcs' ends, so the outline closes. The *tangent* constraint makes a line touch a circle or arc without cutting into it, so there is no corner where they meet. The *equal* constraint keeps both ends the same size, so you set the radius only once.

YOU WILL	Close a roughly drawn slot with coincident, tangent and equal constraints, check that it is fully defined, and find its region.
YOU NEED	Saulitwerx built (Chapter 2). OpenCascade is not needed.
TIME	5 minutes.

1. Run the slot sample:

   TERMINAL

   ```
   cargo run -p swx-sketch --example slot_sketch
   ```

   It prints the slot twice: as drawn, then after solving.

OUTPUT

```
1 as drawn
  left   arc center (  0.00,  0.00)  radius 4.00 mm  from   90° to  260°
  right  arc center ( 28.00,  1.00)  radius 6.00 mm  from  -90° to   90°
  bottom (  0.00, -4.00) -> ( 28.00, -5.00)
  regions: 0
  open edges: [0, 1, 2, 3]
2 solved: true
  dof 0 FullyConstrained
  left   arc center (  0.00,  0.00)  radius 5.00 mm  from   90° to  270°
  right  arc center ( 30.00,  0.00)  radius 5.00 mm  from  -90° to   90°
  bottom (  0.00, -5.00) -> ( 30.00, -5.00)
  regions: 1
    outer edges 4, holes 0, area 378.54 mm²
  open edges: []
```

2. Read stage 1. The slot was drawn carelessly on purpose: the ends have radii 4 and 6, the slot is tilted, and the left arc stops at 260°, 10° short of the bottom side (Figure 5.3). Because of that one gap the outline doesn't close, so there is no region and all four pieces are listed as open edges.
3. Read stage 2. Both radii are now 5 mm, the centers are 30 mm apart and level, and the left arc runs the full half circle from 90° to 270°. The solver turned the arc's end until it met the bottom side.
4. Read `dof 0 FullyConstrained`. Every piece of the slot is pinned down: its size, its shape and where it sits. Nothing is left to chance.
5. Read the region. The closed slot encloses one region of 378.54 mm²: a 30 × 10 mm rectangle plus two half circles of radius 5 mm.

RESULT A fully defined 30 mm slot with 5 mm round ends, and one region ready to become a solid.

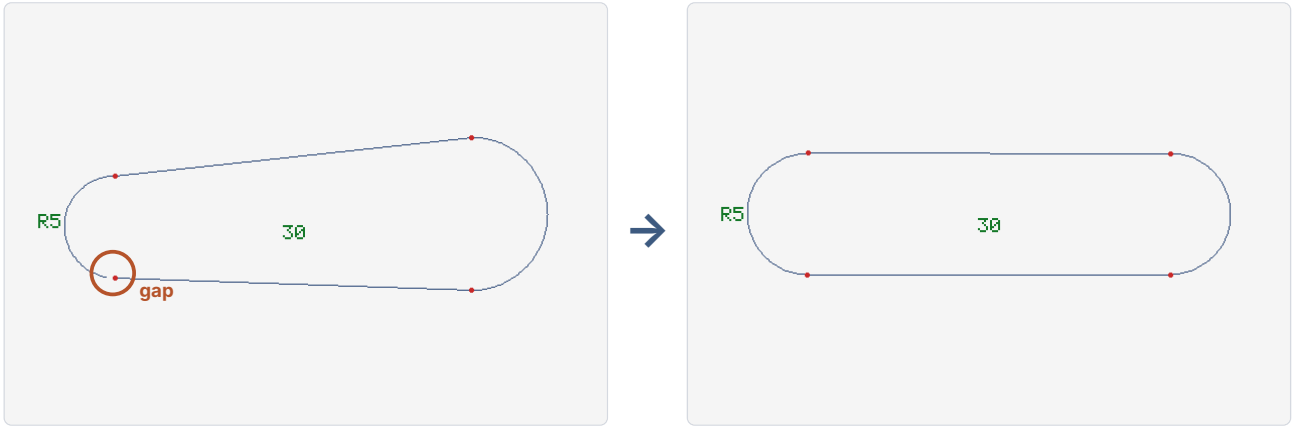


Figure 5.3 The slot before and after solving. As drawn, the ends differ in size, the slot is tilted, and the left arc stops short of the bottom side (ringed). Solved, it is level, with equal 5 mm ends joined all the way round.

How the slot is built

The program is `crates/swx-sketch/examples/slot_sketch.rs`. An arc's two ends are named with `PointRef::ArcStart(arc)` and `PointRef::ArcEnd(arc)`. An arc always runs counterclockwise from its start to its end, so the left arc, drawn from the top (90°) round to the bottom (270°), ends where the bottom side starts:

```
for (a, b) in [
    (PointRef::Start(bottom), PointRef::ArcEnd(left)),
    (PointRef::End(bottom), PointRef::ArcStart(right)),
    (PointRef::Start(top), PointRef::ArcEnd(right)),
```

```
    (PointRef::End(top), PointRef::ArcStart(left)),
] {
    s.add_constraint(Constraint::Coincident(a, b))?;
}
for (line, arc) in [(bottom, left), (bottom, right), (top, left), (top, right)] {
    s.add_constraint(Constraint::Tangent(line, arc))?;
}
s.add_constraint(Constraint::Equal(left, right))?;
s.add_constraint(Constraint::Radius { geom: left, value: 5.0 })?;
```

Then the bottom side is made horizontal, the left center is fixed at the origin, and the centers are set 30 mm apart. That is all the slot needs.

> **TIP** **Which end is which**
>
> Picture walking round the arc counterclockwise, the way the angles grow. Where you set off is `ArcStart`; where you stop is `ArcEnd`. If you join the wrong end, the solver still tries to obey, and you get a different shape from the one you meant. Check the angles in the output, or plot the sketch (Section 9.5).

> **TIP**
>
> A tangent line stays on the side of the circle it started on, and two tangent circles stay touching on the outside or the inside, whichever they were closer to when you added the constraint. Draw roughly the right arrangement and the solver keeps it. When a line's end is joined to an arc's end, tangent there means the line meets the arc square to its radius, which is exactly "no corner".

5.7 Find the region to extrude

Before a sketch can become a solid, Saulitwerx has to know which areas it encloses (Section 4.7). Both samples above end by asking for exactly that. This section shows the call, and how to read the answer.

YOU WILL	Get the regions of a sketch, with their holes and areas, and find the open edges when a sketch doesn't close.
YOU NEED	The two samples above, run once each.
TIME	5 minutes.

1. Look at the end of the centered-plate output (Section 5.5): `regions: 1`, `holes 1`, `area 921.46 mm²`. The rectangle is the region's outside edge and the circle its hole; the area is the rectangle's 1000 mm² minus the hole's 78.54 mm².
2. Look at stage 2 of the slot output (Section 5.6): one region with `outer edges 4` (two lines, two arcs) and no holes.
3. Look at stage 1: `regions: 0` and `open edges: [0, 1, 2, 3]`. The numbers are the pieces in the order they were added to the sketch: 0 and 1 are the arcs, 2 and 3 the lines. When a sketch has no region, the open edges show where to look for the gap.

RESULT You can tell whether a sketch is ready to become a solid, and how much area it encloses. Chapter 7 shows how an outline like this becomes a solid.

How it works

One call returns every region and the open edges:

```rust
use swx_sketch::profile::{profile_report, ProfileOptions};

let profile = profile_report(&s, &ProfileOptions::default())?;
for r in &profile.regions {
    println!("holes {}, area {:.2} mm²", r.holes.len(), r.area());
}
println!("open edges: {:?}", profile.open_edges);
```

`profiles(&s, &options)` returns only the regions. Ends of pieces closer together than a small tolerance count as joined; Section 11.8 lists the setting, every field, and the rules for loops inside loops.

> **CAUTION** **Crossing pieces stop the search**
>
> If two pieces cross, overlap or touch away from their ends, `profile_report` returns an error naming both, such as `geometry 2 and 5 intersect`, and no regions. Saulitwerx doesn't split pieces where they cross. Move or shorten one of them so they meet only end to end.

5.8 Let Saulitwerx add constraints as you draw

Most of a sketch's constraints say what your drawing already almost shows: this corner joins that one, this line is level, this circle touches that line. Adding each of them by hand is slow. *Inference* does it for you. After you add a piece, Saulitwerx compares it with everything already in the sketch, suggests the constraints it nearly satisfies, adds them and solves. It never adds a dimension: sizes stay yours to set.

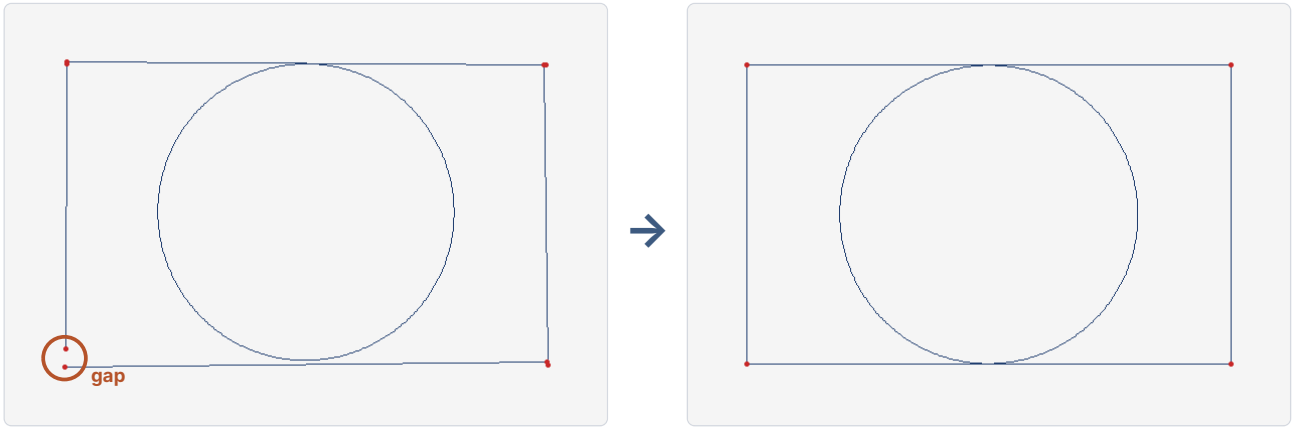


Figure 5.4 The sketch before and after inference. As drawn, the sides lean slightly and the left side stops short of the bottom-left corner (ringed). After inference and solving, the sides are level and upright, every corner is joined, and the circle touches the top and bottom.

YOU WILL	Draw a rough rectangle and a circle one piece at a time, and see which constraints inference adds after each.
YOU NEED	Saulitwerx built (Chapter 2). OpenCascade is not needed.
TIME	5 minutes.

1. Run the inference sample:

   TERMINAL

   ```
   cargo run -p swx-sketch --example inferred_sketch
   ```

   It draws four sides and then a circle, and after each one lists what inference suggested.

OUTPUT

```
bottom: 1 suggested
  horizontal: bottom
right: 2 suggested
  coincident: start of right = end of bottom
  vertical: right
top: 2 suggested
  coincident: start of top = end of right
  horizontal: top
left: 2 suggested
  coincident: start of left = end of top
  vertical: left
  (missed the first corner by 1.31 mm; the default point_tol is 0.047 mm)
again with point_tol 2 mm:
left: 3 suggested
  coincident: start of left = end of top
  coincident: end of left = start of bottom
  vertical: left
corners: (0.13, 0.52) (39.90, 0.52) (39.90, 25.07) (0.13, 25.07)
circle: 2 suggested
  tangent: circle and bottom
  tangent: circle and top
```

2. Read the first three pieces. The bottom was drawn 0.6° off level, so it got horizontal. The right side starts a few hundredths of a millimeter from the bottom's end, so its start was joined to that end, and it got vertical. The top went the same way. No piece was made parallel or perpendicular to another: once a line is horizontal or vertical, those would only repeat the rule.
3. Read the first try at the left side. Its start was joined to the top, but its end stopped 1.31 mm short of the first corner. That is far more than the default reach of 0.047 mm, so that corner was left open.
4. Read the second try. With a reach of 2 mm the gap was closed, and the rectangle is complete. Its corners are now exactly square (Figure 5.4).
5. Read the circle. It was drawn 0.01 mm short of touching the top and bottom, so it got tangent to both.

RESULT A closed rectangle with a circle touching its top and bottom, held by ten constraints you didn't have to add yourself.

Inference doesn't make a sketch fully defined. This rectangle has no sizes yet, and its corner is not fixed, so it still has degrees of freedom (Section 5.3). Add a fixed point and the dimensions next.

How the sample is built

The program is `crates/swx-sketch/examples/inferred_sketch.rs`. After each `add`, one call infers, adds and solves:

```rust
use swx_sketch::infer::{add_inferred, infer, InferOptions};

let right = s.add(Geometry::Line(Line::new(start, end)?))?;
let opts = InferOptions::for_sketch(&s);
let suggested = infer(&s, right, &opts)?;      // look only
let added = add_inferred(&mut s, right, &opts)?; // add and solve
```

`infer` only looks and changes nothing, so you can show the suggestions first and let the user choose. `add_inferred` adds every suggestion and solves. Two settings decide how near is near enough. `point_tol` is the reach for joining points and for tangency; by default it is a thousandth of the sketch's overall size, its diagonal. `angle_tol_deg` is the reach for directions; by default 1°. Section 11.10 gives the full rules.

> **TIP** **Draw from where you stopped**
>
> Inference joins only points that are very close by default. Start each new piece where the last one ended, as precisely as you can, and the corners join themselves. When a drawing is coarser, widen `point_tol`, but not so far that it joins points you meant to keep apart.

> **CAUTION** **It adds what it sees**
>
> A line drawn half a degree off level becomes level, even if you meant it to lean, and a constraint can't be taken out of a sketch yet. When that matters, call `infer` first, and add only the suggestions you want with `add_constraint`. A constraint already in the sketch is never suggested again.

5.9 Common mistakes

- **A value of zero or less.** Distance, radius and diameter must be greater than zero. Adding `value: 0.0` fails with `value must be > 0, got 0`.
- **The wrong kind of point.** `PointRef::Start` and `End` work only on lines, `ArcStart` and `ArcEnd` only on arcs, and `Center` only on circles and arcs. Using `Center` on a line fails with `geometry 0 has a different kind`.
- **Rules that fight each other.** Two fixed points 30 mm apart with a distance of 40 mm between them can't both hold. The solver then stops with `solved: false` and a reason of `Stagnated`, and moves the points as close to meeting the rules as it can. A degree-of-freedom check then says `Conflicting` and names the clashing constraint (Section 5.3).
- **The wrong kind of entity.** Parallel and perpendicular take two lines. Equal takes two lines or two round pieces (circles and arcs), not one of each. Tangent takes a line and a round piece, or two round pieces, never two lines. Each of these fails with `geometry N has a different kind`.
- **An outline with a gap.** Pieces that look joined but aren't leave no region. Join every corner with coincident, including where a line meets an arc (Section 5.6).
- **Expecting inference to set sizes.** Inference adds coincident, horizontal, vertical, parallel, perpendicular and tangent only. Add distances, radii and diameters yourself (Section 5.8).
- **Too few rules.** A sketch can solve and still be under-constrained: the solver keeps unconstrained pieces near where you drew them. Check that the degrees of freedom reach zero before you rely on the sizes.

CHAPTER 6

Solids and STEP files

> **FOR DEVELOPERS**
>
> This chapter calls the geometry engine, `swx-kernel`, from Rust. It needs OpenCascade and the `occt` build feature (Chapter 2).

The part of a CAD program that does the 3D math is its *geometry kernel*. Saulitwerx talks to its kernel through one interface, the `Kernel` trait, so the rest of the program never depends on which kernel is underneath. Today that kernel is OpenCascade, reached through `OcctKernel`.

6.1 Make a solid and save it as STEP

YOU WILL	Build the plate with a hole, measure it, save it as STEP and read it back.
YOU NEED	Saulitwerx built with OpenCascade (Chapter 2).
TIME	5 minutes.

You already ran this program in Chapter 3. Here is what each part of it does. The file is `crates/swx-kernel/examples/plate_with_hole.rs`.

1. **Make the two starting solids.** A box is given by its lowest corner and its three sizes; a cylinder stands on +Z, given by the center of its base, its radius and its height:

```
let plate = k.make_box([0.0, 0.0, 0.0], [40.0, 25.0, 5.0])?;
let tool  = k.make_cylinder([20.0, 12.5, -1.0], 5.0, 7.0)?;
```

2. **Cut one from the other.** The cylinder starts 1 mm below the plate and ends 1 mm above it, so the hole goes cleanly all the way through:

```
let part = k.cut(&plate, &tool)?;
```

3. **Measure the result** with `volume`, `surface_area`, `face_count` and `bounding_box`. The program prints them; Chapter 3 checks them by hand.

4. **Save and read back.** `export_step` writes the file, in millimeters; `import_step` reads it back as one solid. Matching volume and face count show nothing was lost:

```
k.export_step(&part, &out)?;
let back = k.import_step(&out)?;
```

RESULT A STEP file of the plate, checked by reading it back: `wrote plate.step (read back: 4607.30 mm³, 7 faces)`.

> **TIP** **Make the tool overlap**
>
> When you cut a hole, make the tool solid a little longer than the part on both sides. A tool that ends exactly on a face can leave a paper-thin skin, or confuse the kernel.

6.2 Combine solids

The kernel has three Boolean operations (Section 4.9). Each takes two solids and returns a new one; the inputs are left unchanged.

Call	Result
`k.union(&a, &b)`	Everything in `a` or `b`.
`k.cut(&a, &b)`	`a` minus whatever `b` overlaps.
`k.intersect(&a, &b)`	Only what is in both.

> **CAUTION**
>
> Sizes must be finite and greater than zero, and positions finite. A box with a zero size is refused before it reaches the kernel, with `invalid input: box dx must be finite and > 0, got 0`.

6.3 Faces, edges and vertices of a solid

A solid's faces, edges and vertices (Section 4.1) are its *sub-shapes*. Later features will need them: a fillet rounds an edge, a sketch sits on a face. Today you can count them and walk from one to its neighbours. Two sub-shapes are *adjacent* when one bounds the other: a face and the edges around it, or an edge and the two vertices at its ends.

The kernel numbers the sub-shapes of each kind from 0. `SubShape { kind, index }` names one of them, such as face 6, and `TopoKind` is the kind: `Face`, `Edge` or `Vertex`.

YOU WILL	Count the plate's faces, edges and vertices, and find the faces around the hole.
YOU NEED	Saulitwerx built with OpenCascade (Chapter 2).
TIME	5 minutes.

The program is `crates/swx-kernel/examples/part_topology.rs`. It builds the same plate as above, then asks the kernel about its parts.

1. **Count each kind** with `sub_shape_count`:

```rust
for kind in [TopoKind::Face, TopoKind::Edge, TopoKind::Vertex] {
    println!("{kind:?} count: {}", k.sub_shape_count(&part, kind)?);
}
```

   A shared sub-shape counts once: the edge between two faces is one edge.

2. **Go down from a face to its edges** with `adjacent`. Give it the sub-shape to start from and the kind you want back:

```rust
let face = SubShape { kind: TopoKind::Face, index };
let edges = k.adjacent(&part, face, TopoKind::Edge)?;
```

   The list comes back sorted by index, with no repeats.

3. **Go up from an edge to its faces** the same way, asking for `TopoKind::Face`. The program does this for each edge of the hole's wall, the only face with 3 edges.
4. **Run it:**

TERMINAL

```
cargo run -p swx-kernel --features occt --example part_topology
```

You should see:

OUTPUT

```
Face count: 7
Edge count: 15
Vertex count: 10

face 0: 4 edges [0, 1, 2, 3]
face 1: 4 edges [0, 4, 5, 6]
face 2: 5 edges [1, 6, 7, 8, 9]
face 3: 4 edges [2, 7, 10, 11]
face 4: 5 edges [3, 4, 10, 12, 13]
face 5: 4 edges [5, 8, 11, 12]
face 6: 3 edges [9, 13, 14]

hole wall is face 6
  edge 9: faces [2, 6]
  edge 13: faces [4, 6]
  edge 14: faces [6]

face 7: invalid input: face index 7 is out of range: the shape has 7 faces
```

RESULT The plate has 7 faces, 15 edges and 10 vertices. Faces 2 and 4 have 5 edges each, four sides plus the rim of the hole: they are the top and the bottom, the two faces the hole passes through. The hole's wall, face 6, meets them along edges 9 and 13.

> **NOTE** **Why 15 edges, and an edge with one face**
>
> A box has 12 edges; the hole adds its two rims, which makes 14. The fifteenth is a *seam*: the kernel makes a closed round face, like the hole's wall, from a sheet rolled up until its two sides meet, and the line where they meet is an edge. It doesn't show in pictures. Because the wall lies on both sides of it, the seam, edge 14, lists only one face.

> **CAUTION** **A number belongs to one shape**
>
> Indices count the sub-shapes of one shape, so face 6 of this plate means nothing for another shape, even the same plate after a change. Ask again after every operation. An extrude can tell you which of its faces is which (Section 7.5); names that last through changes are on the way, see **Coming next**.

Common mistakes

- **Asking for the same kind you start from,** such as the edges of an edge. That gives: `invalid input: adjacent needs two different kinds, got edge and edge`.
- **Counting from 1.** The last face of a 7-face part is face 6. Face 7 gives: `invalid input: face index 7 is out of range: the shape has 7 faces`.

6.4 Open a STEP file from another program

`import_step` reads any STEP file, and `swx-qa render --step` draws one (Chapter 9). Everything in the file comes in as one shape. Saulitwerx can't edit an imported shape yet; using it as a starting feature is on the roadmap.

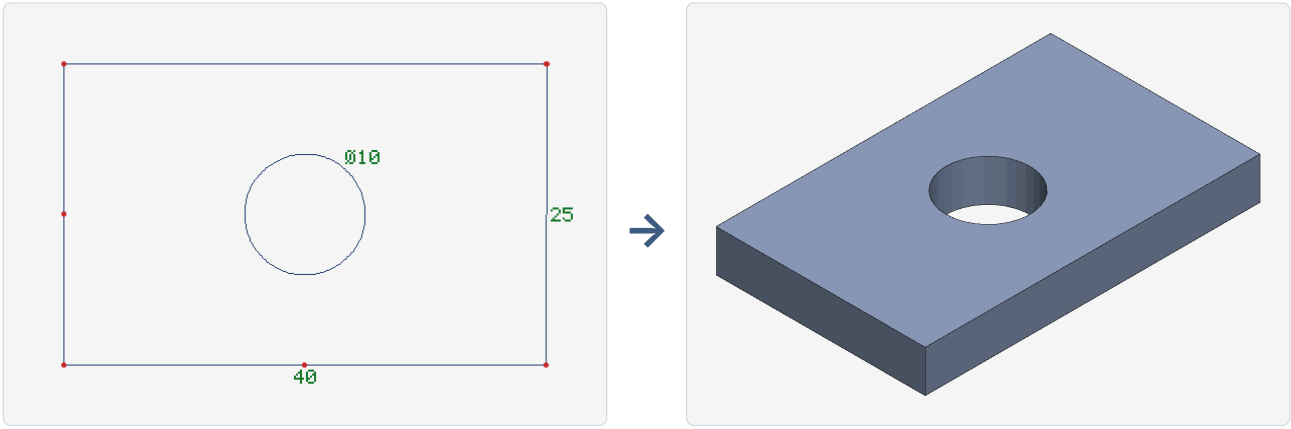
CHAPTER 7

Extruding a profile

> **FOR DEVELOPERS**
>
> This chapter calls the geometry engine, `swx-kernel`, from Rust. It needs OpenCascade and the `occt` build feature (Chapter 2).

An *extrude* takes a flat shape and pushes it straight out into 3D, the way a cookie cutter pressed into dough makes a shape with straight sides. It is the most common way to start a part: draw the outline, then give it a thickness.

Think back to the plate of Chapter 3. There you made it in two steps: a block, then a cylinder cut out of it. With an extrude you draw the plate's outline once, rectangle and hole together, and push it up 5 mm. The hole comes with it, with no separate cut (Figure 7.1).



Profile: rectangle with a hole — Extruded 5 mm

Figure 7.1 An extrude pushes a flat profile straight out. The circle inside the rectangle becomes a hole all the way through.

Two more words come with it:

- The *workplane* is the flat plane the outline is drawn on. It has its own left–right and up–down directions, and a third direction straight out of it, called its *normal*. The extrude goes along the normal.
- The *extent* says how far the extrude goes: a distance one way, or a length split evenly on both sides of the workplane.

In Saulitwerx today: the geometry engine can make a flat face from outlines of lines, arcs and circles on any workplane, and extrude it by a distance or symmetrically. The result can be a new solid, or be joined to a part, cut from it or intersected with it in the same step (Section 7.3). A cut can also go through the whole part, however thick it is (Section 7.4), and a plain extrude can report which face each piece of the outline became (Section 7.5). You type the outline's pieces into a small program; a sketch's regions (Section 5.7) can't be handed to the extrude yet, and the extrude isn't recorded in a part's history.

7.1 Extrude a profile

YOU WILL	Make the plate with a hole from its outline, in one extrude, and save it as STEP.
YOU NEED	Saulitwerx built with OpenCascade (Chapter 2).
TIME	5 minutes.

1. Run the sample program, and tell it to save the part as `extruded-plate.step`:

   TERMINAL
   ```
   cargo run -p swx-kernel --features occt --example extrude_plate -- extruded-plate.step
   ```

   It prints the solid's measurements, then the geometry engine's report about writing the file:

   OUTPUT
   ```
   volume        4607.30 mm³
   faces         7
   bounding box  [0.0, 0.0, 0.0] to [40.0, 25.0, 5.0]
   …  (the geometry engine's file report)
   wrote extruded-plate.step
   ```

2. Compare the numbers with the plate you made in Chapter 3. They are the same: 4607.30 mm³, 7 faces, 40 × 25 × 5 mm. The region's area, 921.46 mm² (Section 5.7), times the 5 mm height gives the volume.

3. Draw the part:

   TERMINAL
   ```
   target/release/swx-qa render --step extruded-plate.step -o extruded-plate.png
   ```

   Open `extruded-plate.png`. You should see the plate of Figure 7.1, with the hole showing light background through it.

RESULT The plate with a hole, made from its outline in one step instead of a block and a cut.

How the sample is built

The program is `crates/swx-kernel/examples/extrude_plate.rs`. It does three things.

1. **Describe each loop.** A `ProfileLoop` is a list of pieces, `Segment2`, in workplane coordinates. Each piece starts where the one before it ended, and the last one ends where the first began. A full circle is a loop on its own:

   ```
   let corners = [[0.0, 0.0], [40.0, 0.0], [40.0, 25.0], [0.0, 25.0]];
   let outer = ProfileLoop {
       segments: (0..4)
           .map(|i| Segment2::Line { start: corners[i], end: corners[(i + 1) % 4] })
           .collect(),
   };
   let hole = ProfileLoop {
       segments: vec![Segment2::Circle { center: [20.0, 12.5], radius: 5.0 }],
   };
   ```

2. **Make the face.** `make_face` takes the workplane, the outer loop, and a list of holes. The hole loops must lie inside the outer one:

   ```
   let face = k.make_face(&Workplane::XY, &outer, &[hole])?;
   ```

3. **Extrude it.** `Extent::Distance(5.0)` pushes the face 5 mm along the workplane's normal, which for the XY plane is up, +Z:

```rust
let part = k.extrude(&face, Extent::Distance(5.0))?;
```

> **TIP** **Direction doesn't matter**
>
> You can list a loop's pieces clockwise or counterclockwise, and any piece may run backwards. Saulitwerx puts them in order before the geometry engine sees them. What matters is that the ends meet: closer than 0.0000001 mm.

7.2 Choose the direction and the workplane

An extrude doesn't have to go up. A negative distance goes the other way, below the workplane. A symmetric extent splits the length evenly, half on each side, which is handy for a part whose middle should sit on the plane, like a bracket centered on its mounting plate.

YOU WILL	Extrude a slot up, down and symmetrically, and see where each result sits.
YOU NEED	Saulitwerx built with OpenCascade.
TIME	5 minutes.

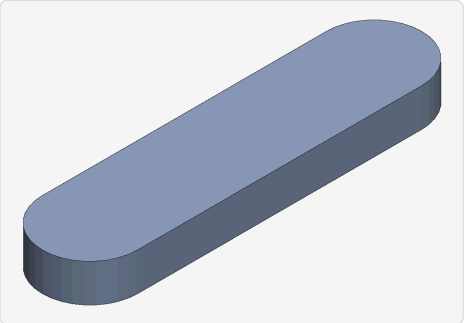


Figure 7.2 The slot you will make: two lines and two half-circles, extruded 4 mm.

1. Run the slot sample:

```
TERMINAL
cargo run -p swx-kernel --features occt --example extrude_slot -- extruded-slot.step
```

It extrudes the slot three ways, 4 mm each, and prints each result's volume and height:

```
OUTPUT
Distance(4.0)   1514.16 mm³, z from 0 to 4
Distance(-4.0)  1514.16 mm³, z from -4 to 0
Symmetric(4.0)  1514.16 mm³, z from -2 to 2
…  (the geometry engine's file report)
wrote extruded-slot.step
```

2. Read the three lines. The volume is the same each time; only the position changes. `Distance(4.0)` sits on the workplane and rises to 4 mm; `Distance(-4.0)` hangs below it; `Symmetric(4.0)` straddles it, 2 mm each side.
3. Draw the first one:

```
TERMINAL
target/release/swx-qa render --step extruded-slot.step -o extruded-slot.png
```

You should see the slot of Figure 7.2: flat sides and round ends.

RESULT You can place an extrude above, below or across its workplane.

The slot's outline mixes lines and arcs. An arc, `Segment2::Arc`, runs counterclockwise from `start_angle` to `end_angle`, in radians measured from the workplane's left–right axis, just as in a sketch (Section 5.6). The right-hand end of the slot is the arc from $-\pi/2$ to $\pi/2$ around (30, 5).

Three workplanes are built in. To draw on another plane, give your own `Workplane` with an `origin`, an `x_dir` and a `normal` at right angles to it (Section 11.5).

Workplane	Left–right	Up–down	Extrude goes
`Workplane::XY`	+X	+Y	+Z
`Workplane::XZ`	+X	+Z	−Y
`Workplane::YZ`	+Y	+Z	+X

CAUTION **XZ extrudes toward −Y**

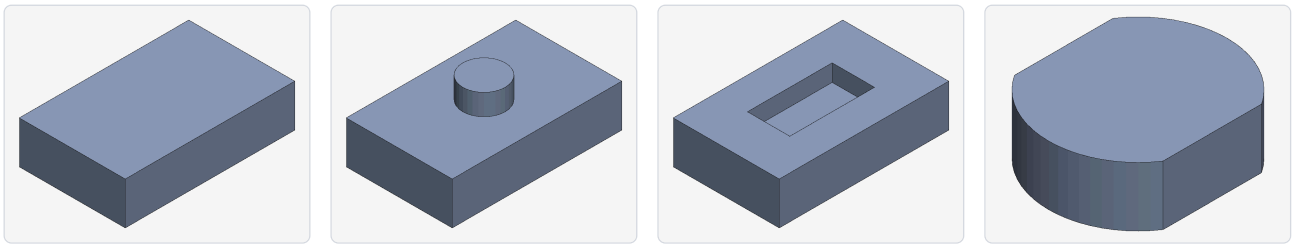
Looking down on the XZ plane so that X runs right and Z runs up, you are looking from the −Y side, and that is where a positive distance goes. If a part comes out on the wrong side, use a negative distance.

7.3 Join, cut or intersect with an extrude

Most extrudes change a part that already exists. A boss stands up from a plate; a pocket sinks into it. You could extrude a new solid and then combine it with a Boolean operation (Chapter 6), but an extrude can do both in one step. Its *extrude mode* says what happens to the extruded shape:

Mode	What you get
New solid	The extruded shape on its own. The part is left as it was.
Join	The part with the extruded shape added to it.
Cut	The part with the extruded shape taken out of it.
Intersect	Only the part's material that lies inside the extruded shape.

The part an extrude joins, cuts or intersects is its *target body*. A good habit: draw the outline on a face of the target, then extrude away from the part to add material, and into it to remove material. Figure 7.3 shows one block and the three combining modes.



Target: 40 × 25 × 10 mm block | Join: circle, 5 mm up | Cut: rectangle, 4 mm down | Intersect: circle, 10 mm down

Figure 7.3 One target and three extrude modes, each outline drawn on the block's top face. The intersect keeps the part of the block inside a 30 mm circle: the block's long sides trim the circle flat. Each picture is scaled to fill its frame.

YOU WILL Add a boss to a block, sink a pocket into it, and trim it to a circle, each with one extrude.
YOU NEED Saulitwerx built with OpenCascade, and `swx-qa` built with it too (Chapter 2).
TIME 10 minutes.

1. Run the sample program, and tell it to save its parts in a folder called `modes`:

   TERMINAL
   ```
   cargo run -p swx-kernel --features occt --example extrude_modes -- modes
   ```

   It prints a line for the block and one for each mode, then two errors on purpose, then the files it wrote:

   OUTPUT
   ```
   target     10000.00 mm³   6 faces  z from 0 to 10
   join       10392.70 mm³   8 faces  z from 0 to 15
   cut         9200.00 mm³  11 faces  z from 0 to 10
   intersect   6505.89 mm³   7 faces  z from 0 to 10
   join, no target -> error: invalid input: extrude mode "join" needs a target body
   cut, all of it -> error: empty result: extrude cut leaves no solid (volume 0e0)
   …  (the geometry engine's file reports)
   wrote modes/modes-target.step
   wrote modes/modes-join.step
   wrote modes/modes-cut.step
   wrote modes/modes-intersect.step
   ```

2. Read the `join` line. The block grew by the boss: a 10 mm circle 5 mm tall adds 392.70 mm³, and the part now reaches 15 mm up.
3. Read the `cut` line. A 20 × 10 mm rectangle cut 4 mm deep removes 800 mm³. The height stays 10 mm, and the pocket's floor and four walls bring the faces to 11.
4. Read the `intersect` line. What is left is the block's material inside a 30 mm circle: less than either one on its own.
5. Draw the cut result:

   TERMINAL
   ```
   target/release/swx-qa render --step modes/modes-cut.step -o modes-cut.png
   ```

   Open `modes-cut.png`. You should see the block with a pocket in its top, as in Figure 7.3. Draw the other files the same way.

RESULT A boss, a pocket and a trimmed block, each made by one extrude against the same target. The block itself is never changed: each mode returns a new solid.

How the sample is built

The program is `crates/swx-kernel/examples/extrude_modes.rs`. The outlines sit on the block's top face, a workplane at z = 10:

```
const TOP: Workplane = Workplane {
    origin: [0.0, 0.0, 10.0],
    x_dir: [1.0, 0.0, 0.0],
    normal: [0.0, 0.0, 1.0],
};
```

Each mode is one call to `extrude_feature`. It takes the face, the extent, the mode and the target. The join goes up, +5 mm, away from the block; the cut goes down, −4 mm, into it:

```
let joined = extrude_feature(&k, &boss, Extent::Distance(5.0),
    ExtrudeMode::Join, Some(&block))?;
let cut = extrude_feature(&k, &pocket, Extent::Distance(-4.0),
    ExtrudeMode::Cut, Some(&block))?;
```

For `ExtrudeMode::NewSolid` the target isn't used, so pass `None`. That gives the same solid as `extrude` (Section 7.1).

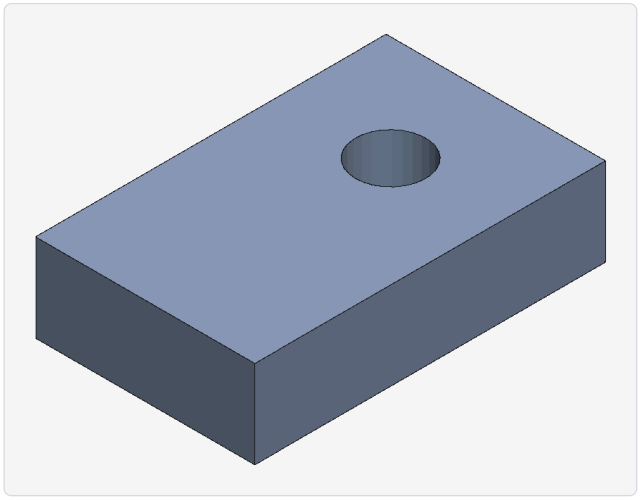
> **TIP** **Which way is "into" the part?**
>
> The extrude goes along the workplane's normal for a positive distance. On a top face whose normal points up, a cut needs a negative distance. If a cut leaves the part unchanged, it probably went the wrong way, into empty space.

> **CAUTION** **A result must have something left**
>
> A cut that removes the whole target, or an intersect whose shapes don't overlap, leaves nothing. That is an error, `empty result: …`, not an empty part. The sample's last line shows one: a rectangle larger than the block, cut all the way through.

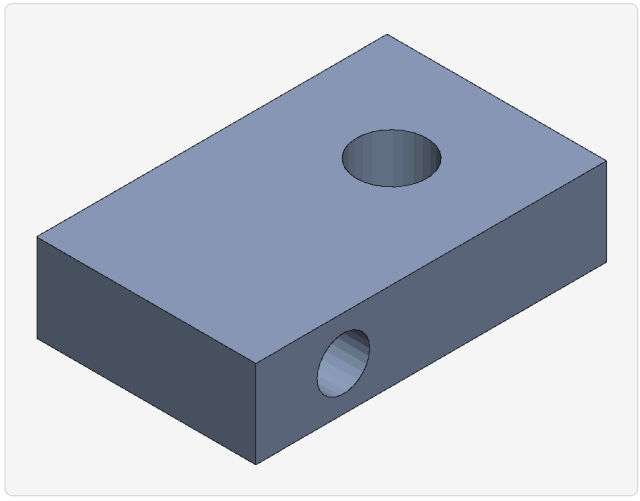
7.4 Cut through the whole part

Holes for bolts and pins usually go all the way through a part. You could measure the part and cut exactly that deep, but if the part later gets thicker, the hole would stop short. A *through-all* cut has no depth of its own: it goes right through the target body, whatever its size.

A through-all cut still needs a direction. It can go along the workplane's normal, against it, or both ways at once. Both ways is the one to use when the outline sits on a plane in the middle of the part (Figure 7.4).



Down from the top face (reversed)



Then both ways from the middle

Figure 7.4 Two through-all cuts in a 40 × 25 × 10 mm block. The first hole is drawn on the top face and cut against its normal, down. The second is drawn on a plane through the middle of the block and cut both ways, out of the front and the back.

YOU WILL	Cut two holes right through a block, one from its top face and one from a plane in its middle.
YOU NEED	Saulitwerx built with OpenCascade, and `swx-qa` built with it too (Chapter 2).
TIME	5 minutes.

1. Run the sample program, and tell it to save its parts in a folder called `through`:

   TERMINAL

   ```
   cargo run -p swx-kernel --features occt --example extrude_through_all -- through
   ```

   It prints a line for the block and one for each cut, then two errors on purpose, then the files it wrote:

OUTPUT

```
block            10000.00 mm³   6 faces
down (reversed)   9497.35 mm³   7 faces
both ways         8790.49 mm³   8 faces
up (normal) -> error: invalid input: the through-all cut misses the target: it removes nothing
join -> error: invalid input: a through-all extent works only with the cut mode, not "join"
…  (the geometry engine's file reports)
wrote through/through-down.step
wrote through/through-both.step
```

2. Read the `down` line. An 8 mm hole through 10 mm of block removes 502.65 mm³, and adds one face: the hole's wall.
3. Read the `both ways` line. A 6 mm hole through the block's 25 mm width removes another 706.86 mm³. The cut started in the middle of the block and went out through both sides.
4. Read the first error. From the top face, the normal points up, away from the block. A through-all cut that way finds nothing to remove, and says so.
5. Draw the result with both holes:

TERMINAL

```
target/release/swx-qa render --step through/through-both.step -o through-both.png
```

Open `through-both.png`. You should see the block of Figure 7.4, with a hole down through its top and a smaller one through its side.

RESULT Two holes that go right through the block. If the block were 50 mm thick instead of 10, the same two cuts would still go through it.

How the sample is built

The program is `crates/swx-kernel/examples/extrude_through_all.rs`. The extent is `Extent::ThroughAll`, with a direction:

Direction	The cut goes
`ThroughDirection::Normal`	Along the workplane's normal.
`ThroughDirection::Reversed`	Against the normal.
`ThroughDirection::Both`	Both ways from the workplane.

The first hole is drawn on the top face, so it goes against the normal, down into the block:

```
let drilled = extrude_feature(&k, &top_hole,
    Extent::ThroughAll(ThroughDirection::Reversed),
    ExtrudeMode::Cut, Some(&block))?;
```

The second is drawn on a workplane through the middle of the block, at y = 12.5, and cut from the result of the first:

```
let cross = extrude_feature(&k, &cross_hole,
    Extent::ThroughAll(ThroughDirection::Both),
    ExtrudeMode::Cut, Some(&drilled))?;
```

NOTE **How far is “all”?**

Saulitwerx measures the outline and the target, and makes the cut a little longer than the distance needed to clear the whole target. That is why a through-all extent works only with `extrude_feature` and the cut mode: it needs a target to measure. `extrude` on its own refuses it.

TIP **A missed cut is an error**

If a through-all cut removes nothing, Saulitwerx refuses it instead of handing back the part unchanged, so a cut that went the wrong way never passes unnoticed. Change the direction, or move the outline over the part.

7.5 Find the faces an extrude made

Most features after the first one point at something already in the part: round *this* edge, draw on *that* face, cut down to *this* face. The kernel numbers every face and edge (Section 6.3), but the numbers come from the geometry engine, not from you, so you can’t tell in advance which number the top of the plate gets.

An extrude can tell you. Ask for it *with history*, and next to the solid you get a record of where each face came from. Faces of an extrude come in two sorts:

- The two *caps*: the flat faces at each end of the push. On the plate, the bottom and the top.
- One *side face* for each edge of the outline: the wall it swept as it moved. A straight edge sweeps a flat wall; the hole’s circle sweeps a round one.

Follow one piece of your outline through both steps, and you know its face: the rectangle’s line at x = 40 became edge 1 of the flat face, and edge 1 swept face 1 of the plate (Figure 7.5).

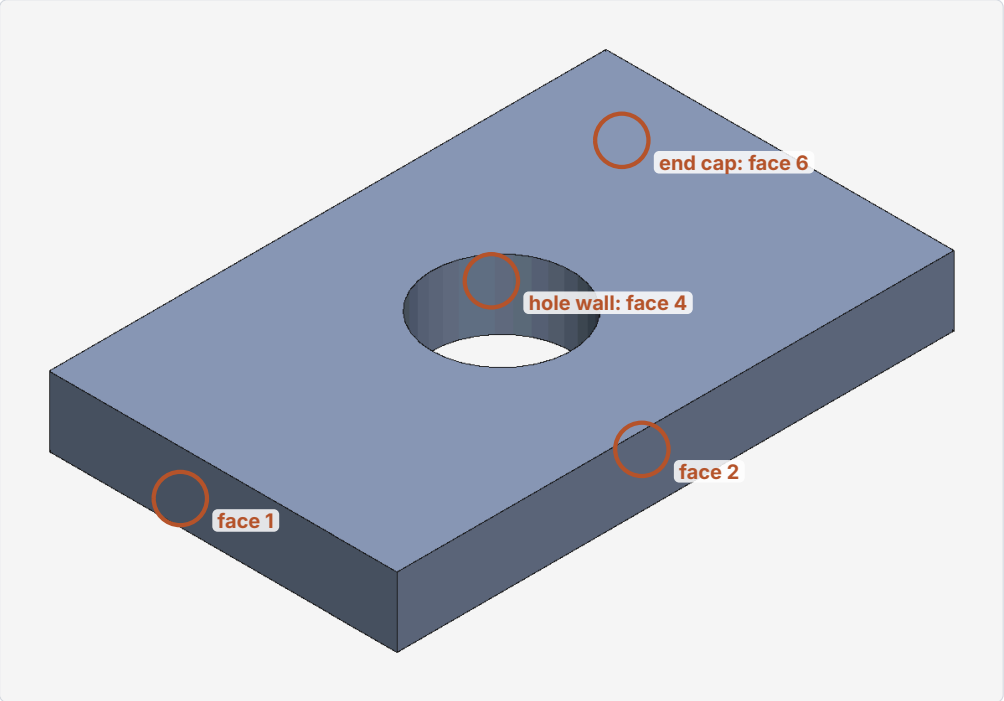


Figure 7.5 The plate’s faces, as its extrude history names them. Face 1 was swept by the rectangle’s line at x = 40, face 2 by its line at y = 25, face 4 by the hole’s circle. The start cap, face 5, is underneath.

In Saulitwerx today: the geometry engine keeps this record for a plain extrude, the one that makes a new solid. You get it in your own program. The extrude with join, cut or intersect doesn't give one yet, the part's history doesn't keep it, and no later feature can use it to find a face yet.

YOU WILL	Extrude the plate with history, and find which face each piece of its outline became.
YOU NEED	Saulitwerx built with OpenCascade (Chapter 2).
TIME	5 minutes.

1. Run the sample program:

   TERMINAL
   ```
   cargo run -p swx-kernel --features occt --example extrude_history
   ```

   It prints the two caps, then one line for each piece of the outline, then where the top is:

   OUTPUT
   ```
   start cap     face 5
   end cap       face 6
   outer 0  ->  edge 0  ->  face 0   x 0..40  y 0..0  z 0..5
   outer 1  ->  edge 1  ->  face 1   x 40..40  y 0..25  z 0..5
   outer 2  ->  edge 2  ->  face 2   x 0..40  y 25..25  z 0..5
   outer 3  ->  edge 3  ->  face 3   x 0..0  y 0..25  z 0..5
   hole 0  ->  edge 4  ->  face 4   x 15..25  y 7.5..17.5  z 0..5
   end cap is    x 0..40  y 0..25  z 5..5
   ```

2. Read the line `outer 1`. The second line of the rectangle, from (40, 0) to (40, 25), became edge 1 of the face, and edge 1 swept face 1. The box after it shows where that face is: flat at x = 40, the full 25 mm wide and 5 mm tall. That is the short side at the front left in Figure 7.5.
3. Read the line `hole 0`. The circle swept face 4, the hole's wall: 10 mm across and 5 mm tall, around the hole's center.
4. Read the last line. The end cap lies flat at z = 5 mm: it is the top of the plate.

RESULT Every face of the plate, named by the piece of the outline that made it. Faces 0 to 4 are the sides, 5 the bottom and 6 the top.

How the sample is built

The program is `crates/swx-kernel/examples/extrude_history.rs`. It uses the outline of `extrude_plate`, and three calls do the work.

1. **Make the face with history.** `make_face_tracked` takes the same inputs as `make_face`. It also returns, for each loop (the outer one first, then the holes), the edge each piece became, in the order you listed the pieces:

   ```rust
   let (face, edges) = k.make_face_tracked(&Workplane::XY, &outer, &[hole])?;
   ```

   Here `edges[0][1]` is the edge made from the outer loop's second piece.

2. **Extrude it with history.** `extrude_tracked` takes the same inputs as `extrude`, and returns an `ExtrudeHistory` with the solid:

   ```rust
   let (part, history) = k.extrude_tracked(&face, Extent::Distance(5.0))?;
   ```

   `history.start_cap` and `history.end_cap` are the two caps. `history.sides` holds one pair per edge of the face: the edge, then the side face it swept.

3. **Follow an edge to its face**, and ask where that face is. `sub_shape_bounding_box` gives the smallest axis-aligned box around one face, edge or vertex:

```
let side = history.sides.iter()
    .find(|(e, _)| e == edge)
    .map(|(_, f)| *f)
    .ok_or("edge not in history")?;
let bb = k.sub_shape_bounding_box(&part, side)?;
```

> **CAUTION** **The start cap isn't always the face you drew**
>
> The sweep runs from the start cap to the end cap, along the workplane's normal. With a positive distance, the start cap is where your outline was drawn. With a negative one, the sweep starts below it: the end cap is the face you drew, and the start cap is the far end. With a symmetric extent, neither cap is on the workplane. When it matters, check with `sub_shape_bounding_box`, as the sample does.

> **NOTE** **Same plate, other numbers**
>
> The plate of Section 6.3, made as a block with a cylinder cut out, has its hole's wall as face 6. The extruded plate has it as face 4. The two shapes are the same; the steps that made them differ, and so do the numbers. That is why a number is only good for the shape it came from, and why the history is worth having: it tells you which number is which.

> **TIP** **Your order, not the engine's**
>
> However you list a loop's pieces, clockwise or counterclockwise, the edges come back in your order: the second piece you gave is always `edges[loop][1]`. You don't have to know how Saulitwerx reordered the loop.

7.6 Common mistakes

- **The ends don't quite meet.** A loop whose last piece stops short of the first is refused with `outer loop: consecutive segments don't meet within 0.0000001`. Compute shared corners once and reuse them, as `extrude_plate` does with `corners`.
- **A circle mixed with other pieces.** A full circle must be a loop of its own. To round a corner, use an arc.
- **A hole outside the outline, or touching its edge.** The geometry engine refuses the face: `make_face failed: the face is invalid …`. Keep every hole wholly inside the outer loop.
- **An outline that crosses itself,** like a figure eight. Same message; or, when the two halves are the same size, `outer loop encloses no area`, because their areas cancel out. Split it into separate outlines.
- **A distance of zero.** An extrude has to go somewhere: `extrude distance must be finite and non-zero, got 0`.
- **Join, cut or intersect with no target.** These modes need a part to work on: `invalid input: extrude mode "cut" needs a target body`. Pass the part as `Some(&part)`, or use `ExtrudeMode::NewSolid`.
- **A cut that goes the wrong way.** It misses the part, and the part comes back unchanged. Check the sign of the distance (Section 7.3).
- **A cut that removes everything,** or an intersect that doesn't overlap: `empty result: extrude cut leaves no solid …`. Make the outline smaller than the face it cuts, or move it over the part.
- **A through-all extent with another mode.** Only a cut can go through all: `a through-all extent works only with the cut mode, not "join"`. Give a distance for a join or an intersect.

- **A through-all extent with history.** `extrude_tracked` has no target to measure, so it refuses through-all just as `extrude` does: `a through-all extent needs a target body to measure; use extrude_feature with the cut mode`.
- **A face number from another shape.** The numbers in an extrude history belong to that solid. After a cut, face 6 may be another face, or not exist: `sub_shape_bounding_box` then gives the wrong face's box, or `face index 7 is out of range: the shape has 7 faces` (Section 6.3). Take the numbers again from the step that made the shape you ask about.
- **A through-all cut the wrong way.** It misses the part: `the through-all cut misses the target: it removes nothing`. On a top face, use `ThroughDirection::Reversed`, or `Both` (Section 7.4).

CHAPTER 8

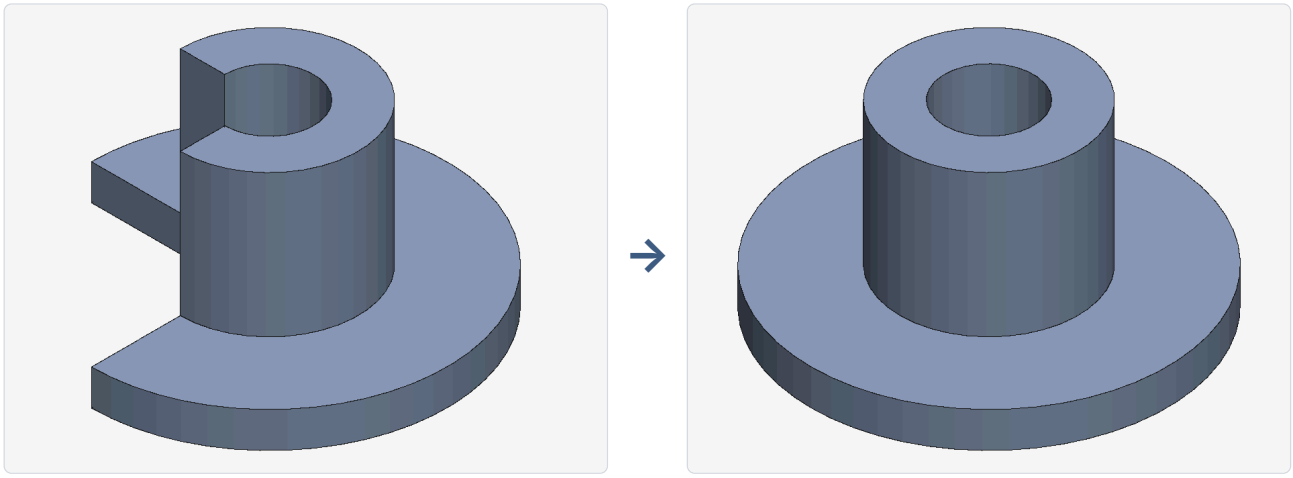
Revolving a profile

> **FOR DEVELOPERS**
>
> This chapter calls the geometry engine, `swx-kernel`, from Rust. It needs OpenCascade and the `occt` build feature (Chapter 2).

A *revolve* turns a flat shape around a line, the way a potter's wheel turns a lump of clay into a pot. Every point of the shape sweeps a circle around that line, so the result is round: a shaft, a knob, a washer, a bottle.

Where an extrude (Chapter 7) pushes an outline straight out, a revolve swings it around. You draw only half of the part's cross-section, the half on one side of the center line, and the revolve makes the rest.

The line the shape turns around is the *axis of revolution*. The *revolve angle* says how far it turns: a full turn makes a closed, round part; less than a full turn makes a wedge, like a slice of a cake.



Three quarters of a turn A full turn

Figure 8.1 A flanged bushing made by one revolve. Stopped at three quarters of a turn (left), the open quarter shows the outline that was turned: an L-shape beside a hollow center. A full turn (right) closes the part.

In Saulitwerx today: the geometry engine can revolve a flat face from `make_face` about any axis, by any angle up to a full turn. Like an extrude, the result can be a new solid, or be joined to a part, cut from it or intersected with it in the same step. Before it turns anything, it checks that the axis lies in the outline's plane and stays clear of the outline, and says plainly when it doesn't (Section 8.3). You type the outline's pieces into a small program; a sketch's regions can't be handed to the revolve yet, and the revolve isn't recorded in a part's history.

8.1 Revolve a profile

YOU WILL	Make a flanged bushing from half its cross-section, with a full turn and with three quarters of a turn.
YOU NEED	Saulitwerx built with OpenCascade, and `swx-qa` built with it too (Chapter 2).
TIME	10 minutes.

1. Run the sample program, and tell it to save its parts in a folder called `revolve`:

   TERMINAL
   ```
   cargo run -p swx-kernel --features occt --example revolve_bushing -- revolve
   ```

   It prints a line for each part, then five errors on purpose, then the files it wrote. The two longest error lines are broken here to fit the page; on your screen each is one line:

   OUTPUT
   ```
   full turn   8482.30 mm³   6 faces  x from -20 to 20
   270°        6361.73 mm³   8 faces  x from -20 to 20
   grooved     8362.92 mm³  10 faces  x from -20 to 20
   360 -> error: invalid input: revolve angle must be in (0, 2π] radians, got 360
   zero axis -> error: invalid input: revolve axis direction must be non-zero
   cut, no target -> error: invalid input: revolve mode "cut" needs a target body
   axis through the outline -> error: invalid input: the profile crosses
     the revolve axis (origin [12.0, 0.0, 0.0], direction [0.0, 0.0, 1.0]):
     it reaches 7 on one side and 8 on the other
   axis out of the plane -> error: invalid input: revolve axis (origin
     [0.0, 0.0, 0.0], direction [0.0, 1.0, 0.0]) is not in the profile's
     plane (through [0.0, 0.0, 0.0], normal [0.0, -1.0, 0.0])
   …  (the geometry engine's file reports)
   wrote revolve/bushing.step
   wrote revolve/bushing-270.step
   wrote revolve/bushing-grooved.step
   ```

2. Read the `full turn` line. The outline reaches 20 mm from the axis, so the part runs from x = −20 to 20 mm: 40 mm across. The six faces are the bore, the tube's outside, the flange's rim, and the top, the flange's top and the bottom.

3. Read the `270°` line. Three quarters of a turn gives three quarters of the volume, 6361.73 mm³. The two flat ends of the wedge add two faces.

4. Draw the three-quarter part:

   TERMINAL
   ```
   target/release/swx-qa render --step revolve/bushing-270.step -o bushing-270.png
   ```

   Open `bushing-270.png`. You should see the left picture of Figure 8.1: the open quarter shows the L-shaped outline that was turned. Draw `revolve/bushing.step` the same way to see the full part.

RESULT A round part made from one flat outline. The bushing's bore, tube and flange came from a single six-sided shape.

How the sample is built

The program is `crates/swx-kernel/examples/revolve_bushing.rs`. It does three things.

1. **Draw half the cross-section.** The outline sits on the XZ workplane, so left–right is X, the distance from the axis, and up–down is Z, the height. It starts 5 mm from the axis, which leaves a 10 mm bore down the middle:

```
let outline = polygon(&[
    [5.0, 0.0], [20.0, 0.0], [20.0, 4.0],
    [10.0, 4.0], [10.0, 20.0], [5.0, 20.0],
]);
let section = k.make_face(&Workplane::XZ, &outline, &[])?;
```

`polygon` is a small helper in the sample that joins the points with lines, the last back to the first.

2. **Give the axis.** An `Axis` is a point on the line and the line's direction. Here it is the Z axis, through the origin and pointing up:

```
let z_axis = Axis { origin: [0.0, 0.0, 0.0], dir: [0.0, 0.0, 1.0] };
```

3. **Revolve it.** `revolve_feature` takes the face, the axis, the angle in radians, the mode and the target. `TAU` is a full turn, 2π; `1.5 * PI` is three quarters of one:

```
let bushing = revolve_feature(&k, &section, z_axis, TAU,
    ExtrudeMode::NewSolid, None)?;
let cutaway = revolve_feature(&k, &section, z_axis, 1.5 * PI,
    ExtrudeMode::NewSolid, None)?;
```

NOTE **How close is close enough?**

The checks allow 0.0000001 mm (`REVOLVE_AXIS_TOLERANCE`), the same small gap the geometry engine treats as zero. An axis that misses the plane by less than that still counts as in it, and an outline that pokes past the axis by less than that still counts as touching it.

TIP **Which way does it turn?**

Point your right thumb along the axis direction: your fingers curl the way a positive angle turns. Here the axis points up, so the outline swings from +X towards +Y, counterclockwise seen from above. A full turn looks the same either way.

CAUTION **Radians, not degrees**

The geometry engine takes the angle in radians: a full turn is 2π, about 6.283, and 360 is refused. To work in degrees, multiply by π/180, or write `360.0_f64.to_radians()` .

8.2 Cut or join with a revolve

A revolve uses the same four modes as an extrude (Section 7.3): new solid, join, cut and intersect, with a target body for the last three. A revolve cut is how you make a round groove, such as the seat for a sealing ring around a shaft (Figure 8.2).

YOU WILL	Cut a groove around the bushing with a second revolve.
YOU NEED	The sample run of Section 8.1.
TIME	5 minutes.

1. Read the `grooved` line of the output in Section 8.1. The groove took 119.38 mm³ from the bushing, and its floor and two walls added three faces, bringing the bushing's six faces to nine, plus one more where the groove splits the tube's outside in two: ten.
2. Draw the grooved part:

   TERMINAL

   ```
   target/release/swx-qa render --step revolve/bushing-grooved.step -o bushing-grooved.png
   ```

   Open `bushing-grooved.png`. You should see Figure 8.2: a narrow groove around the tube, halfway up.

RESULT A groove around a round part, made by turning a small rectangle around the same axis as the part itself.

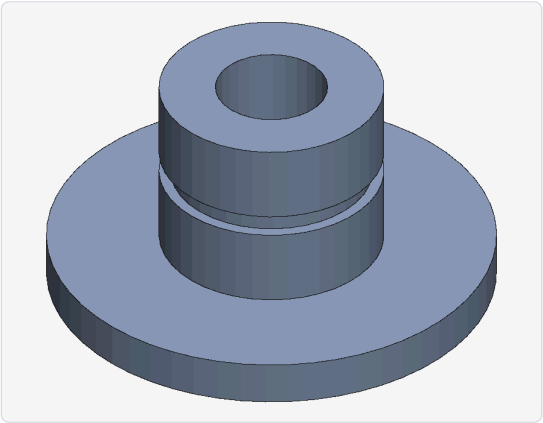


Figure 8.2 The bushing with a groove around its tube: a small rectangle, revolved a full turn and cut from the part.

The groove's outline is a 2 × 2 mm square that straddles the tube's outer wall, at x = 10 mm. Only the half inside the tube removes material; the other half turns through empty space. Letting a cutting outline overlap the edge like this makes sure the groove opens cleanly onto the surface:

```
let groove = k.make_face(&Workplane::XZ,
    &polygon(&[[9.0, 11.0], [11.0, 11.0], [11.0, 13.0], [9.0, 13.0]]), &[])?;
let grooved = revolve_feature(&k, &groove, z_axis, TAU,
    ExtrudeMode::Cut, Some(&bushing))?;
```

The rules are the extrude's: join, cut and intersect need a target, a result with nothing left is an `empty result` error, and the target itself is never changed. `ExtrudeMode` is the name of the mode type for both features.

8.3 Common mistakes

- **Degrees instead of radians.** `revolve angle must be in (0, 2π] radians, got 360`. Convert the angle (Section 8.1).
- **An angle of zero, or a negative one.** A revolve has to turn: the same message. To turn the other way, reverse the axis direction.
- **An axis with no direction.** `revolve axis direction must be non-zero`. Give the axis a direction, such as `[0.0, 0.0, 1.0]`.
- **The axis runs through the outline.** The outline would sweep through itself, so the revolve is refused: `the profile crosses the revolve axis (…): it reaches 7 on one side and 8 on the other`. The two numbers say how far, in millimeters, the outline reaches on each side; the sample's flange runs from 5 to 20 mm, and an axis at 12 mm splits it 7 and 8. Move the axis, or the outline, so the whole outline is on one side. Touching the axis is fine, and makes a solid center with no bore.

- **The axis isn't in the outline's plane.** An axis that is tilted out of the plane, stands straight out of it, or runs beside it is refused: `revolve axis (…) is not in the profile's plane (through …, normal …)`. The message gives a point on the plane and its normal, the direction straight out of it; the axis direction must be at right angles to that normal, and the axis point must lie on the plane. Draw the outline on a plane that contains the axis, as the sample does with the XZ plane and the Z axis.
- **A shape that isn't one flat face.** `revolve needs a single planar face`. Give the revolve a face from `make_face`, not a solid.
- **Join, cut or intersect with no target.** `revolve mode "cut" needs a target body`. Pass the part as `Some(&part)`, or use `ExtrudeMode::NewSolid`.

CHAPTER 9

Pictures of parts and sketches

`swx-qa render` draws a part to a PNG picture file, and `swx-qa sketch` draws a sketch. They need no graphics card and no window, and the same input always gives the same picture, byte for byte. The project uses them to show its work in reviews; this manual's pictures come from them too.

9.1 Draw a STEP file

YOU WILL Draw any STEP file to a PNG, at the size you want.
YOU NEED `swx-qa` built with OpenCascade (Chapter 2), and a STEP file. `plate.step` from Chapter 3 works.

1. Run `render` with the STEP file, a size and an output name:

   TERMINAL

   ```
   target/release/swx-qa render --step plate.step --size 1600x1000 -o plate.png
   ```

   It answers `wrote plate.png (1600x1000, …)`, with the file size in bytes.

2. Open `plate.png`. The part fills the picture, seen from the same corner as every other render.

RESULT A PNG of the part, ready to share or to put in a document.

The renderer chooses how finely to break curved faces into flat pieces for you: the mesh stays within 0.1 % of the part's overall size of the true surface.

9.2 Choose the size

`--size` takes the width and height in pixels, written `W` `x` `H`, for example `1600x1000`. The default is `800x600`; the largest is `8192x8192`. For a picture you will print, allow about 80 pixels per centimeter (200 per inch) of printed width: a picture printed 10 cm wide needs about 800 pixels.

9.3 Choose which edges are outlined

The renderer outlines the edges where faces meet at a sharp angle. `--crease` sets how sharp, in degrees, from 0 to 180; the default is 30. Where the surface bends by more than this angle from one face to the next, the edge gets a dark line. The edges of a box bend by 90°, so they are always outlined; the many small facets of a curved face bend by only a few degrees, so they are not. A smaller value outlines more edges; a larger value outlines fewer.

> **TIP**
>
> On a curved part that looks striped with lines, raise `--crease` (try 45). If an edge you want to see has no outline, lower it.

9.4 Draw a built-in sample

`--fixture` draws one of three built-in parts instead of a file: `cube` (10 mm), `cylinder` (5 mm radius, 10 mm tall) or `l-bracket` (30 mm legs, 8 mm thick, 20 mm deep). They need no OpenCascade, so they are a quick way to test the tool.

TERMINAL

```
target/release/swx-qa render --fixture l-bracket -o bracket.png
```

Section 11.1 lists every option.

9.5 Draw a sketch

`swx-qa sketch` solves a built-in sketch and draws it from above, with x to the right and y up. Lines, circles and arcs are dark blue; points and the ends of lines are red dots; each dimension's value is written in green beside what it measures.

YOU WILL	Draw a built-in sketch to a PNG.
YOU NEED	`swx-qa` built (Chapter 2). OpenCascade is not needed.
TIME	2 minutes.

1. Run `sketch` with a sample name and an output name:

   TERMINAL

   ```
   target/release/swx-qa sketch --fixture rectangle -o rectangle.png
   ```

   It answers `wrote rectangle.png (800x600, …)`, with the file size in bytes.

2. Open `rectangle.png`. You should see a 40 × 25 mm rectangle, labeled `40` along the bottom and `25` up the right side.

3. Try the other sample, a right triangle with sides of 4 and 3:

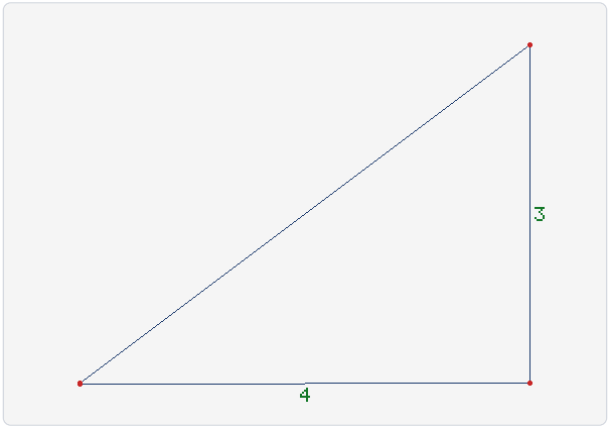
   TERMINAL

   ```
   target/release/swx-qa sketch --fixture triangle-345 -o triangle.png
   ```

RESULT A PNG of a solved sketch with its dimension values (Figure 9.1).



rectangle



triangle-345

Figure 9.1 The two built-in sketches. The triangle's long side has no number: it has no dimension of its own, and comes out 5 because the other two sides are 4 and 3.

`--size` works as it does for parts; the default is `800x600`. The sketch is scaled to fill the picture with a margin, so a small sketch and a large one come out the same size. Labels are drawn at a fixed size in pixels, so they look smaller in a large picture.

FOR DEVELOPERS

To draw your own sketch, call `swx_qa::sketch::plot(&sketch, &PlotOptions::default())` and save the result with `to_png()`. It draws the sketch as it is, solved or not. The manual's sketch pictures come from `crates/swx-qa/examples/plate_sketch_plot.rs`, which does exactly this:

TERMINAL

```
cargo run -p swx-qa --example plate_sketch_plot -- <dir> 600x420
```

It writes `sketch-rough.png`, `sketch-shape.png` and `sketch-sized.png` in `<dir>`.

Only distances, radii and diameters are labeled. Other constraints, such as horizontal or fixed, don't show in the picture, and there are no dimension arrows yet.

CHAPTER 10

Parameters and history

> **FOR DEVELOPERS**
>
> This chapter uses the document model, `swx-core`, and the sketcher, `swx-sketch`, from Rust. It needs no OpenCascade.

The document model holds what makes a part parametric: its named parameters and its ordered list of features (Section 4.8, Section 4.10). It doesn't make geometry itself. Its job is to keep the history valid and to work out what must be rebuilt after a change.

10.1 See what a change rebuilds

YOU WILL	Build a four-feature history and ask which features a change affects.
YOU NEED	Saulitwerx built (Chapter 2).
TIME	5 minutes.

1. Run the sample:

   TERMINAL

   ```
   cargo run -p swx-core --example feature_history
   ```

   OUTPUT

   ```
   history:
     0  Sketch1
     1  Extrude1
     2  Sketch2
     3  Revolve1
   change thickness -> rebuild Extrude1
   edit Sketch2     -> rebuild Sketch2, Revolve1
   add Extrude2 with depth `height` -> error: unknown parameter `height`
   ```

2. Read the history. It is the table in Section 4.10: two sketches, an extrude that uses `thickness`, and a revolve that uses `sweep`.
3. Read the two rebuild lines. Changing `thickness` touches only Extrude1, the one feature that uses it. Editing Sketch2 rebuilds Sketch2 and Revolve1, which consumes it; Sketch1 and Extrude1 are left alone.
4. Read the last line. A feature may use only parameters that exist, so adding an extrude whose depth is `height` is refused.

RESULT You saw Saulitwerx decide, from the history alone, the smallest set of features a change needs rebuilt.

10.2 How the sample is built

The program is `crates/swx-core/examples/feature_history.rs`. A `Document` holds parameters and features:

```
let mut doc = Document::new();
doc.set_param("thickness", 5.0);
let plate_sketch = doc.add_feature("Sketch1", FeatureKind::Sketch)?;
doc.add_feature("Extrude1", FeatureKind::Extrude {
    profile: plate_sketch,
    depth: "thickness".into(),
})?;
```

Then two questions give the rebuild lists, in history order:

```
doc.dirty_after_param_change("thickness");   // [Extrude1]
doc.dirty_after_feature_edit(pin_sketch)?;    // [Sketch2, Revolve1]
```

The features that need rebuilding are called *dirty*. Today's feature kinds are `Sketch`, `Extrude` (with a depth parameter) and `Revolve` (with an angle parameter, in degrees). They are records in the history only; making their geometry comes with the part features on the roadmap.

10.3 Name and rename parameters

Every part keeps its parameters in one parameter table. Each row has a name, a value, a unit and an optional comment, and says where the parameter came from. A *model parameter* belongs to a dimension and gets the next free name in the series `d0`, `d1`, `d2`, … A *user parameter* is one you create yourself, with a name you choose, for a value you want to reuse or change in one place.

Names like `d0` are quick to make but say nothing. Renaming `d0` to `thickness` makes the part easier to read, and every feature that uses it follows the new name, so nothing breaks.

YOU WILL	Build a small parameter table, rename a parameter a feature uses, and see which names are refused.
YOU NEED	Saulitwerx built (Chapter 2).
TIME	5 minutes.

1. Run the sample:

TERMINAL

```
cargo run -p swx-core --example parameter_table
```

OUTPUT

```
model parameters: d0, d1, d2
after renaming d0 to thickness:
  name       kind   value  unit  comment
  thickness  model      5  mm
  d1         model     40  mm
  d2         model     10  mm
  clearance  user     0.2  mm    gap around the pin
Extrude1 depth reads `thickness`
next model parameter: d3
add `2nd_hole` -> error: invalid parameter name `2nd_hole`: must start with a letter or `_`
and contain only letters, digits and `_`
rename d1 to thickness -> error: a parameter named `thickness` already exists
```

2. Read the first line. Three dimensions of a plate (its thickness, width and hole size) were given the model names `d0`, `d1` and `d2`, in the order they were made.
3. Read the table. `d0` is now called `thickness`; it is still a model parameter. `clearance` is a user parameter, with a comment saying what it is for. The rows stay in the order they were made.
4. Read the line about Extrude1. The extrude was made with depth `d0`. After the rename it reads `thickness`, with no change to the extrude itself.
5. Read `next model parameter: d3`. The name `d0` became free when it was renamed, but it is not handed out again. A model name always means one parameter, for the life of the part.
6. Read the last two lines. A name may not start with a digit, and two parameters may not share a name. In both cases the table is left as it was.

RESULT You have a parameter table with meaningful names, and a feature that kept working through a rename.

> **TIP Name what you will change**
>
> Rename the parameters you expect to edit later, such as a thickness or a hole size. Leave the rest as `d0`, `d1`, ...

How the sample is built

The program is `crates/swx-core/examples/parameter_table.rs`. Each call works on the document:

```
let depth = doc.add_model_param(5.0, "mm");        // "d0"
doc.add_user_param("clearance", 0.2, "mm")?;
doc.set_param_comment("clearance", "gap around the pin")?;
doc.rename_param(&depth, "thickness")?;
for p in doc.parameters() { /* name, kind, value, unit, comment */ }
```

The unit is free text, such as `"mm"` or `"deg"`, and may be empty; Saulitwerx doesn't check it yet. `set_param`, used in the history sample above, changes a parameter's value; given a new name, it makes a user parameter with no unit. Section 11.11 lists every call.

A parameter can also hold a formula instead of a plain number, so that it follows other parameters (Section 10.6).

10.4 Drive a sketch with parameters

A sketch dimension can be tied to a parameter, so the parameter table becomes the place where you change the part's sizes. Each tied dimension makes its own model parameter, `d0`, `d1`, ... in the order you add them. Change the parameter, and the dimension follows the next time you apply the table and solve. The tie is called a *binding*.

YOU WILL	Bind a rectangle's width and height to parameters, rename one, change it, and see the sketch follow.
YOU NEED	Saulitwerx built (Chapter 2). OpenCascade is not needed.
TIME	5 minutes.

1. Run the sample:

```
TERMINAL
cargo run -p swx-sketch --example driven_plate
```

OUTPUT

```
dimension 40 mm -> parameter d0
dimension 25 mm -> parameter d1
solved: 40.00 × 25.00 mm
renamed d0 to width (1 dimension follows)
width = 60: 60.00 × 25.00 mm
rename d1 without rename_bound -> error: a sketch dimension is bound to unknown parameter `d1`
height = 0 -> error: value must be > 0, got 0
sketch still 60.00 × 25.00 mm
```

2. Read the first two lines. The width dimension, 40 mm, made parameter `d0`; the height, 25 mm, made `d1`.
3. Read `renamed d0 to width`. The parameter got a meaningful name, and the sketch was told, so its width dimension now reads `width`.
4. Read `width = 60`. Setting `width` to 60 and applying the table made the rectangle 60 mm wide. The height stayed 25 mm.
5. Read the last three lines. Renaming `d1` in the table alone left the sketch looking for `d1`, which no longer exists. A height of 0 was refused, and the sketch kept its size.

RESULT A rectangle whose size you change from the parameter table.

How the sample is built

The program is `crates/swx-sketch/examples/driven_plate.rs`. Instead of `add_constraint`, a dimension goes in with `bind_dimension`, which also adds its parameter:

```rust
use swx_sketch::params::{apply_params, bind_dimension, rename_bound};

let (_, name) = bind_dimension(&mut s, &mut doc, Constraint::Distance {
    a: PointRef::Start(bottom), b: PointRef::End(bottom), value: 40.0 })?;
// name == "d0"
doc.rename_param("d0", "width")?;
rename_bound(&mut s, "d0", "width");
doc.set_param("width", 60.0);
apply_params(&mut s, &doc)?;              // copy the values into the sketch
s.solve(SolveOptions::default())?;        // then move the geometry
```

CAUTION **Rename in both places**

The document and the sketch keep their own lists. `doc.rename_param` renames the parameter and the features that use it, but not the sketch's bindings. Call `rename_bound` with the same two names straight after, or `apply_params` fails with `a sketch dimension is bound to unknown parameter`.

TIP

Only distance, radius and diameter can be bound. A diameter's parameter holds the diameter, not the radius. `apply_params` checks every value before it changes anything, so one bad value leaves the whole sketch as it was.

10.5 Work out an expression

An expression lets one size depend on others: a hole that is always half the plate's width plus 5 mm is `d1 / 2 + 5 mm`. Saulitwerx reads the formula, looks up each name, and checks the units as it goes. Adding a length to an angle is an error, but a length times a length is an area.

YOU WILL	Work out six expressions from a small parameter table, and see each error an expression can give.
YOU NEED	Saulitwerx built (Chapter 2).
TIME	5 minutes.

1. Run the sample:

   TERMINAL

   ```
   cargo run -p swx-core --example expressions
   ```

   The table holds `d0` = 5 mm, `d1` = 40 mm, `wall` = 2.5 mm and `angle` = 30 deg.

   OUTPUT

   ```
   d1 / 2 + 5 mm        = 25.00 mm
   (d1 - 2 * wall) / 3  = 11.67 mm
   2 in + d0            = 55.80 mm
   90 deg - angle       = 60.00 deg
   d1 * d0              = 200.00 mm²
   d1 / d0              = 8.00 (unitless)
   d1 + 5 deg
     error: dimension mismatch at byte 3: cannot apply `+` to length and angle
   5 mmm
     error: unknown unit `mmm` at byte 2 (expected mm, cm, m, in, ft, deg or rad)
   d9 * 2
     error: unknown name `d9` at byte 0
   d1 / (d0 - 5 mm)
     error: division by zero at byte 3
   d1 / (2
     error: syntax error at byte 7: expected `)`
   ```

2. Read the first line. Half of 40 mm is 20 mm, plus 5 mm is 25 mm. Multiplication and division come before addition and subtraction, as in school arithmetic; brackets change the order, as in the second line.
3. Read `2 in + d0`. Units can be mixed: 2 inches is 50.8 mm, plus 5 mm. Results are always worked out in millimeters and radians.
4. Read the next three lines. An angle minus an angle is an angle. A length times a length is an area, in mm². A length divided by a length is a plain number with no unit.
5. Read the errors. Each one says what is wrong and where, counting characters from 0: in `d1 + 5 deg`, byte 3 is the `+`.

RESULT You can work out a size from other sizes, in any of seven units, and trust that mixed-up units are caught.

How the sample is built

The program is `crates/swx-core/examples/expressions.rs`. `parse` reads the text once; `eval` works it out, asking a function you supply for each name's value:

```rust
use swx_core::expr::{eval, parse, Quantity, Unit};

let resolve = |name: &str| -> Option<Quantity> {
    let p = doc.parameters().iter().find(|p| p.name == name)?;
    let unit = Unit::from_suffix(&p.unit)?;
    Some(Quantity::new(p.value * unit.factor(), unit.dim()))
};
let q = eval(&parse("d1 / 2 + 5 mm")?, &resolve, Unit::Mm)?;
// q.value == 25.0 (mm), q.dim == Dim::LENGTH
```

The parameter table stores a unit as plain text, so the sample turns it into a `Unit` itself. Section 11.12 lists the rules and every unit.

> **TIP** **Give every number its unit**
>
> `d1 + 5` is an error, because `d1` is a length and `5` is a bare number. Write `d1 + 5 mm`. A bare number is right only where it scales something, as in `d1 / 2`.

10.6 Make one size follow another

A parameter can hold an expression instead of a plain number. Then it is worked out from the parameters it names, and it changes whenever they do. Give a plate's hole the formula `d0 / 2 + 5 mm`, and when you make the plate wider, the hole grows with it. You change one number; the rest follow.

Saulitwerx works the formulas out in *dependency order*: every parameter after the ones its formula uses. It also refuses a formula that would make a loop, such as `a` from `b` while `b` comes from `a`, because a loop has no answer.

YOU WILL	Give two parameters formulas, change and rename the one they depend on, and see which formulas are refused.
YOU NEED	Saulitwerx built (Chapter 2).
TIME	5 minutes.

1. Run the sample:

TERMINAL

```
cargo run -p swx-core --example parameter_formulas
```

The table starts with `d0` = 40 mm, the plate's width, and `wall` = 2.5 mm. `d1` gets the formula `d0 / 2 + 5 mm` and `d2` gets `(d0 - 2 * wall) / 3`.

OUTPUT

```
  name   value  unit  formula
  d0     40.00  mm
  d1     25.00  mm    d0 / 2 + 5 mm
  d2     11.67  mm    (d0 - 2 * wall) / 3
  wall    2.50  mm
worked out in order: d0, d1, wall, d2
change d0 -> rebuild Extrude1
after d0 = 60:
  name   value  unit  formula
  d0     60.00  mm
  d1     35.00  mm    d0 / 2 + 5 mm
  d2     18.33  mm    (d0 - 2 * wall) / 3
  wall    2.50  mm
after renaming d0 to width:
  name   value  unit  formula
  width  60.00  mm
  d1     35.00  mm    width / 2 + 5 mm
  d2     18.33  mm    (width - 2 * wall) / 3
  wall    2.50  mm
width = d1 - 5 mm -> error: parameter expressions form a cycle: width -> d1 -> width
d1 = width * wall -> error: expression for `d1` gives length^2, but its unit expects length
d1 = height / 2 -> error: expression for `d1`: unknown name `height` at byte 0
after d1 = 30 (typed as a plain number):
  name   value  unit  formula
```

```
width  60.00  mm
d1     30.00  mm
d2     18.33  mm    (width - 2 * wall) / 3
wall    2.50  mm
```

2. Read the first table. Each parameter with a formula shows it, and its value is the formula's result: 40 / 2 + 5 = 25 mm, and (40 − 5) / 3 = 11.67 mm.
3. Read `worked out in order`. `d2` needs `wall`, which was made after it, so `d2` is worked out last. Parameters that need nothing keep the order they were made in.
4. Read `change d0 -> rebuild Extrude1`. The sample's one extrude has depth `d1`, and `d1` comes from `d0`, so changing `d0` rebuilds the extrude, though the extrude never names `d0` (Section 4.10).
5. Read the table after `d0 = 60`. `d1` and `d2` followed on their own: 35 mm and 18.33 mm.
6. Read the table after the rename. The formulas now say `width`, and every value is the same.
7. Read the three errors. `width` from `d1` would make a loop, so it is refused, and the message traces it. `width * wall` is an area, which can't go in a length. `height` doesn't exist. Each time the table is left exactly as it was.
8. Read the last table. Typing a plain number into `d1` replaced its formula. `d1` is 30 mm from now on, and no longer follows `width`.

RESULT A parameter table where one change updates every size that depends on it.

How the sample is built

The program is `crates/swx-core/examples/parameter_formulas.rs`. `set_param_expr` stores a formula and works out the whole table:

```rust
let width = doc.add_model_param(40.0, "mm");      // d0
let hole = doc.add_model_param(0.0, "mm");        // d1
doc.add_user_param("wall", 2.5, "mm")?;
doc.set_param_expr(&hole, "d0 / 2 + 5 mm")?;    // d1 is now 25
doc.set_param(&width, 60.0);                      // d1 follows: 35
doc.evaluation_order();                           // names, dependencies first
```

Each `Parameter` has an `expr` field holding the formula as typed, or `None` for a plain number. The result is stored in the parameter's own unit: a formula giving 1 inch, in a parameter whose unit is `mm`, stores 25.4. A bare number, such as `5`, is taken in the parameter's unit. Section 11.11 lists every call.

> **CAUTION** **A plain number replaces the formula**
>
> `set_param` stores a number and drops any formula the parameter had. To change how a parameter is worked out, give it a new formula with `set_param_expr` instead.

> **TIP** **When a change can't be worked out**
>
> If a new value would break a formula that depends on it, such as a count of 0 under `10 mm / n`, the value is still stored, but the formulas that depend on it keep their old results without an error. Check the values after such a change.

10.7 Common mistakes

- **Using a parameter before it exists.** Call `set_param` first; otherwise `add_feature` fails with `unknown parameter`.
- **Naming a parameter badly.** A name starts with a letter or `_` and holds only letters, digits and `_`: `hole_2` works, `2nd_hole` and `hole-2` don't. Names are case-sensitive.
- **Reusing a name.** Every parameter needs its own name, whether it is a model or a user parameter. To swap two names, rename one of them to a spare name first.
- **Renaming a bound parameter in one place only.** Call `rename_bound` after `rename_param` (Section 10.4).
- **Forgetting to apply and solve.** Changing a parameter changes the table only. Call `apply_params`, then `solve`, to move the sketch.
- **A formula that loops.** A parameter can't depend on itself, directly or through others. Break the loop by making one of them a plain number.
- **Replacing a formula by accident.** `set_param` drops the formula (Section 10.6).
- **Using a feature that comes later.** A feature may use only features already in the history. Otherwise `add_feature` fails with `feature N references feature M which is not earlier in the history`.

PART IV

Reference

Every command, option, operation and setting that exists today, with its default and its allowed values.

CHAPTER 11

Reference

11.1 `swx-qa render`

Draws a part as an isometric, flat-shaded PNG with its sharp edges outlined.

FORM

```
swx-qa render (--fixture <name> | --step <file.step>) -o <out.png> [options]
```

Option	What it does	Default
`--fixture <name>`	Draws a built-in part: `cube`, `cylinder` or `l-bracket`. Give this or `--step`, not both.	—
`--step <file>`	Draws a STEP file. Needs a build with `--features occt`.	—
`-o`, `--output <file>`	The PNG to write. Required. An existing file is replaced.	—
`--size <W>x<H>`	Picture size in pixels. Each side from 1 to 8192.	`800x600`
`--crease <deg>`	Outline edges where the surface bends by more than this many degrees. 0 to 180.	`30`
`-h`, `--help`	Prints the help text and does nothing else.	—

On success it prints `wrote <file> (<W>x<H>, <bytes> bytes)` and exits with status 0. On an error it prints `swx-qa:` and the message (Section 12.2) and exits with status 1, without writing a file.

Built-in part	Shape
`cube`	A 10 mm cube with one corner at the origin.
`cylinder`	Radius 5 mm, height 10 mm, standing on +Z; 32 flat segments around.
`l-bracket`	An L profile with 30 mm legs and 8 mm thickness, 20 mm deep.

11.2 `swx-qa sketch`

Solves a built-in sketch and draws it from above as a PNG, with its dimension values.

FORM

```
swx-qa sketch --fixture <name> -o <out.png> [--size <W>x<H>]
```

Option	What it does	Default
`--fixture <name>`	The built-in sketch to draw: `rectangle` or `triangle-345`. Required.	—
`-o`, `--output <file>`	The PNG to write. Required. An existing file is replaced.	—
`--size <W>x<H>`	Picture size in pixels. Each side from 1 to 8192.	`800x600`
`-h`, `--help`	Prints the help text and does nothing else.	—

Success and errors work as for `render`. `--step` and `--crease` belong to `render` only.

Built-in sketch	Shape
`rectangle`	40 × 25 mm, lower-left corner fixed at the origin; sides horizontal and vertical; bottom and right dimensioned.
`triangle-345`	A right triangle with its right angle fixed at the origin; a horizontal side of 4 and a vertical side of 3, both dimensioned. The long side comes out 5.

In the picture	Drawn as
Lines, circles, arcs	Dark blue lines, 1 pixel wide.
Points, line ends	Red dots, 5 pixels across.
Distance	Its value, in green, beside the middle of the two points, on the side away from the sketch's center.
Radius, diameter	Its value with `R` or `Ø` in front, outside the circle at the upper right, or outside an arc at its middle.
Values	Up to 3 decimals, with trailing zeros dropped: `40`, `12.5`, `R2.125`.

11.3 Build options

Command	What it builds
`cargo test --workspace`	Everything without OpenCascade, and runs every test.
`cargo build --release -p swx-qa --features occt`	The `swx-qa` tool with STEP input, at `target/release/swx-qa`.
`cargo run -p swx-sketch --example plate_sketch`	Runs the sample sketch (Chapter 5).
`cargo run -p swx-sketch --example sketch_dof`	Runs the degree-of-freedom sample (Section 5.3).
`cargo run -p swx-sketch --example centered_plate -- [<width> <height>]`	Runs the centered-hole sample (Section 5.5). Defaults: 40 and 25.
`cargo run -p swx-sketch --example slot_sketch`	Runs the slot sample (Section 5.6).
`cargo run -p swx-qa --example plate_sketch_plot -- <dir> [<W>x<H>]`	Draws the sample sketch in three stages, as PNGs in `<dir>` (Section 9.5).
`cargo run -p swx-qa --example constraint_plots -- <dir> [<W>x<H>]`	Draws the centered-plate and slot sketches as PNGs in `<dir>`.
`cargo run -p swx-kernel --features occt --example plate_with_hole -- <file.step>`	Builds the plate and saves it (Chapter 6). The file defaults to `plate.step`.
`cargo run -p swx-kernel --features occt --example part_topology`	Counts the plate's faces, edges and vertices and lists their neighbours (Section 6.3).
`cargo run -p swx-kernel --features occt --example plate_stages -- <dir>`	Saves the block, the cylinder and the plate as three STEP files in `<dir>`.
`cargo run -p swx-kernel --features occt --example extrude_plate -- <file.step>`	Extrudes the plate from its outline (Section 7.1).
`cargo run -p swx-kernel --features occt --example extrude_slot -- <file.step>`	Extrudes a slot three ways (Section 7.2).
`cargo run -p swx-kernel --features occt --example extrude_modes -- <dir>`	Joins, cuts and intersects extrudes with a block (Section 7.3). STEP files in `<dir>`.
`cargo run -p swx-kernel --features occt --example extrude_through_all -- <dir>`	Cuts two holes through a block (Section 7.4). STEP files in `<dir>`.
`cargo run -p swx-kernel --features occt --example extrude_history`	Extrudes the plate with history and prints which face each piece of the outline became (Section 7.5).

Command	What it builds
`cargo run -p swx-kernel --features occt --example revolve_bushing -- <dir>`	Revolves a bushing and cuts a groove in it (Chapter 8). STEP files in `<dir>`.
`cargo run -p swx-core --example feature_history`	Runs the history sample (Chapter 10).
`cargo run -p swx-core --example parameter_table`	Runs the parameter-table sample (Section 10.3).
`cargo run -p swx-sketch --example driven_plate`	Runs the bound-dimension sample (Section 10.4).
`cargo run -p swx-core --example expressions`	Runs the expression sample (Section 10.5).
`cargo run -p swx-core --example parameter_formulas`	Runs the stored-formula sample (Section 10.6).
`cargo run -p swx-sketch --example inferred_sketch`	Runs the inference sample (Section 5.8).
`cargo run -p swx-qa --example inference_plots -- <dir> [<W>x<H>]`	Draws the inference sketch as drawn and after inference, as PNGs in `<dir>`.

The `occt` feature switches on OpenCascade. It is off by default, so a plain build needs only Rust.

11.4 Geometry kernel operations

Operation	What it does and what it accepts
`make_box(corner, size)`	A box with its lowest corner at `corner` and sides `size` = [x, y, z]. Sizes > 0.
`make_cylinder(base, r, h)`	A cylinder along +Z with its base circle centered at `base`. Radius and height > 0.
`make_sphere(center, r)`	A sphere. Radius > 0.
`union(a, b)`, `cut(a, b)`, `intersect(a, b)`	Boolean operations (Section 4.9). Inputs unchanged.
`volume(s)`, `surface_area(s)`	In mm³ and mm².
`bounding_box(s)`	The smallest box, aligned with the axes, that holds the solid: its `min` and `max` corners.
`face_count(s)`	The number of faces.
`sub_shape_count(s, kind)`	The number of faces, edges or vertices (`kind`), each shared one counted once. Their indices run from 0 to the count minus 1 (Section 6.3).
`adjacent(s, of, kind)`	The sub-shapes of `kind` that touch `of`, sorted by index, without repeats: going down, the edges or vertices of a face, or the vertices of an edge; going up, the faces or edges that contain `of`. `kind` must differ from `of.kind`, and `of.index` must be in range.
`sub_shape_bounding_box(s, of)`	The smallest axis-aligned box around one face, edge or vertex `of` of `s`, as for `bounding_box`. `of.index` must be in range.
`tessellate(s, deflection)`	Breaks the surface into triangles, no farther than `deflection` (mm, > 0) from the true surface.
`export_step(s, path)`	Writes a STEP file in millimeters, replacing any existing file.
`import_step(path)`	Reads everything in a STEP file as one shape.

All positions must be finite numbers; all sizes finite and greater than zero.

Type	Fields and rules
`TopoKind`	`Face`, `Edge` or `Vertex`.
`SubShape { kind, index }`	One face, edge or vertex: the `index`-th of its `kind`, counted from 0. Valid only for the shape it came from; ask again after each operation.

11.5 Profiles, extrude and revolve

`make_face(plane, outer, holes)` makes a flat face on `plane`, bounded by the `outer` loop, with each loop in `holes` cut out. `extrude(face, extent)` sweeps that face along the plane's normal into a new solid (Chapter 7).

Type	Fields and rules
`Workplane`	`origin` (a point), `x_dir` (the plane's left–right direction) and `normal` (straight out of the plane). Both directions non-zero and at right angles; their length doesn't matter. A 2D point (u, v) sits at `origin + u·x_dir + v·(normal × x_dir)`.
`Workplane::XY`, `::XZ`, `::YZ`	Built-in planes through the origin, with normals +Z, −Y and +X (Section 7.2).
`Segment2::Line { start, end }`	A straight piece. Its ends must be apart.
`Segment2::Arc { center, radius, start_angle, end_angle }`	A circular arc, counterclockwise from `start_angle` to `end_angle`, in radians from the +u axis. Radius > 0; the two angles must give different end points.
`Segment2::Circle { center, radius }`	A full circle. Radius > 0. It must be the only piece of its loop.
`ProfileLoop { segments }`	Pieces in chain order, each meeting the next and the last meeting the first. Any piece may run backwards, and the loop may go either way round.
`Extent::Distance(d)`	`d` mm along the normal; negative goes the other way. Not zero.
`Extent::Symmetric(l)`	`l` mm in total, half on each side of the plane. Not zero.
`Extent::ThroughAll(dir)`	Through the whole target body (Section 7.4). Only `extrude_feature` with `ExtrudeMode::Cut` accepts it; `extrude` refuses it.
`ThroughDirection::Normal`, `::Reversed`, `::Both`	Which way a through-all cut goes: along the normal, against it, or both ways.
`LOOP_GAP_TOLERANCE`	1e-7 mm: the largest gap between two ends that still counts as joined. Ends that close are moved to meet exactly.
`PERPENDICULAR_TOLERANCE`	1e-9: how far from a right angle `x_dir` and `normal` may be.
`REVOLVE_AXIS_TOLERANCE`	1e-7 mm: how far a revolve's axis may be off the face's plane, and how far the face may reach past the axis, and still pass (Section 8.3).

All numbers must be finite. A face is refused if a loop is empty, has no area, doesn't close, or crosses itself, or if a hole isn't wholly inside the outer loop. Every refusal is an error (Section 12.2), never a crash.

`make_face_tracked(plane, outer, holes)` and `extrude_tracked(face, extent)` do the same as `make_face` and `extrude`, and also say where each piece went (Section 7.5). They take the same inputs and give the same errors; `extrude_tracked` refuses `Extent::ThroughAll` as `extrude` does.

Returns	What it holds
`make_face_tracked` ⟶ `(face, edges)`	`edges[l][i]`: the edge of `face` made from piece `i` of loop `l`, with loop 0 the outer loop and loop 1 onward the holes in order. Pieces in the order you listed them, whichever way round the loop goes.
`extrude_tracked` ⟶ `(solid, history)`	The solid, as `extrude` makes it, and its `ExtrudeHistory`.
`ExtrudeHistory::start_cap`	The face the sweep starts from: the input face moved by the extent's start, a face of the solid. For a positive distance it is on the workplane; for a negative distance or a symmetric extent it isn't.
`ExtrudeHistory::end_cap`	The face where the sweep ends: the start cap moved by the length along the normal.
`ExtrudeHistory::sides`	One `(edge, face)` pair per edge of the input face, in edge index order: an edge of the input face, then the side face of the solid it swept.

`extrude_feature(k, face, extent, mode, target)` extrudes `face` like `extrude`, then combines the result with `target` as `mode` says (Section 7.3). It returns a new solid; `target` is left unchanged.

Mode or value	Result and rules
`ExtrudeMode::NewSolid`	The extrusion alone. `target` is ignored and may be `None`.
`ExtrudeMode::Join`	Target ∪ extrusion. Needs a target.
`ExtrudeMode::Cut`	Target − extrusion. Needs a target.
`ExtrudeMode::Intersect`	Target ∩ extrusion. Needs a target.
`EMPTY_VOLUME_TOLERANCE`	1e-9 mm^3: a result with less volume than this counts as empty and is refused with `EmptyResult`.

A through-all extent with any mode but cut, and a missing target, are checked first, before any geometry is made. A through-all cut is made a little longer than the distance from the outline to the far side of the target, measured from their bounding boxes. If it leaves the target's volume unchanged (within a relative 1e-9), it is refused as a miss. Errors from `extrude` and from the Boolean operations come through unchanged.

`revolve(face, axis, angle)` turns a face from `make_face` about `axis` by `angle` radians into a new solid (Chapter 8). `revolve_feature(k, face, axis, angle, mode, target)` revolves it and combines the result with `target`, with the same modes and rules as `extrude_feature` (Section 8.2).

Type or value	Fields and rules
`Axis { origin, dir }`	The line through the point `origin` along `dir`. `origin` finite; `dir` finite and non-zero, any length (it is scaled to length 1).
`angle`	Radians, greater than 0 and at most 2π (a full turn; up to 1e-12 over 2π also counts as a full turn). A positive angle turns counterclockwise looking against `dir`, from its tip (right-hand rule).

The axis and the angle are checked before any geometry is made. Then `revolve` checks the face against the axis: the face must be one flat face, the axis must lie in its plane, and the face must not have points on both sides of the axis. Each check allows `REVOLVE_AXIS_TOLERANCE`, 1e-7 mm; a face that touches the axis passes (Section 8.3).

11.6 Sketch entities and constraints

Entity	Defined by
`Point`	x, y.
`Line`	A start point and an end point.
`Circle`	A center point and a radius > 0.
`Arc`	A center, a radius > 0, and start and end angles in radians, running counter-clockwise; the end angle must be after the start. Solving can change all five; it keeps the arc's sweep more than 0 and at most one full turn.

Constraint	Rule
`Coincident(p, q)`	Points `p` and `q` are at the same place.
`Horizontal(line)`	The line's two ends are at the same height.
`HorizontalPoints(p, q)`	`p` and `q` are at the same height.
`Vertical(line)`	The line's two ends are directly above each other.
`VerticalPoints(p, q)`	`p` is directly above or below `q`.
`Fixed { point, x, y }`	The point stays at (x, y).
`Distance { a, b, value }`	Points `a` and `b` are `value` apart. Value > 0.
`Radius { geom, value }`	The circle or arc has this radius. Value > 0.

Constraint	Rule
`Diameter { geom, value }`	The circle or arc has this diameter. Value > 0.
`Parallel(l1, l2)`	Lines `l1` and `l2` point the same way (or exactly opposite ways).
`Perpendicular(l1, l2)`	Lines `l1` and `l2` are at 90°.
`Equal(g1, g2)`	Two lines of equal length, or two circles or arcs of equal radius (any mix of circle and arc). A line and a circle can't be made equal.
`Midpoint { point, line }`	The point is halfway between the line's two ends.
`Symmetric { a, b, line }`	Points `a` and `b` are mirror images across the line: the middle of `a` – `b` lies on the line (or on its extension), and `a` – `b` is perpendicular to it.
`Tangent(g1, g2)`	A line and a circle or arc, in either order, touch: the distance from the center to the line (extended without end) equals the radius. Or two circles or arcs touch, from outside or inside.

A point `p` is written `PointRef::Point(id)` for a point entity, `Start(id)` or `End(id)` for a line's ends, `Center(id)` for a circle's or arc's center, or `ArcStart(id)` or `ArcEnd(id)` for the point of an arc at its start or end angle. Every constraint that takes a point accepts all six.

Tangent detail	How it is decided
Line and circle	The circle stays on the side of the line its center was on when the constraint was added.
Two circles	Touching from outside (centers `r1 + r2` apart) or inside (`|r1 – r2|` apart), whichever the drawing was closer to when the constraint was added.
Line end joined to arc end	When a coincident joins one of the line's ends directly to one of the arc's ends, tangent means the line is square to the arc's radius at that shared point. This holds whichever of the two constraints is added first; a join made through a separate point entity is not recognized.
Where they touch	Otherwise anywhere along the line. Tangent alone doesn't say where a line stops; add other constraints for its ends.

Tangency uses the standard distance conditions for lines and circles; see A. Bowyer and J. Woodwark, *A Programmer's Geometry*, chapters 3 and 5.

11.7 Solver settings and report

Setting	Meaning	Default
`tolerance`	Done when the remaining error is at most this.	`1e-10`
`max_iterations`	Give up after this many attempts.	`100`
`initial_damping`	How cautious the first steps are.	`1e-6`

Stop reason	Meaning
`Converged`	Every rule holds, within the tolerance. `converged` is `true`.
`MaxIterations`	The attempts ran out first.
`Stagnated`	No move makes it better: usually rules that conflict.
`NonFinite`	A value became infinite or not a number.

The solver uses the Levenberg–Marquardt method, a standard way to solve many equations at once; see J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed., chapter 10.

11.8 Profiles

`swx_sketch::profile::profile_report(&sketch, &options)` finds the regions a sketch encloses and the pieces left open (Section 4.7). `profiles(&sketch, &options)` returns the regions alone. Point entities are ignored.

Setting	Meaning	Default
`weld_tolerance`	Ends of pieces closer than this count as one corner. `None` uses 0.000001 × the diagonal of the box around the sketch. Must be > 0 when given.	`None`

Result	Meaning
`regions`	Every region found. Each has an `outer` loop and a list of `holes`.
`Region::area()`	The region's area: the outer loop's area minus its holes', in mm².
`Loop`	The `edges` around it in order, each a piece's id and whether it is walked backwards (`reversed`), and its `signed_area`.
`signed_area`	Positive for an outer loop, which runs counter-clockwise; negative for a hole, which runs clockwise. Exact for arcs.
`open_edges`	Ids of the pieces in no loop: free-ended pieces, pieces that only link two loops, and pieces of zero length. Smallest id first.

Rule	What happens
Circle	A loop on its own. So is an arc that goes all the way round.
Loop inside a loop	Becomes a hole of the smallest loop around it. A loop inside that hole is a new region (an island), and so on, alternating.
Pieces that cross, overlap or touch away from their ends	An error, `SelfIntersecting`, naming the first two such pieces. No regions are returned.
No closed loop	No regions, and no error.

Loops are found by walking around each corner in angle order, the standard method for the faces of a flat network of lines; see M. de Berg, O. Cheong, M. van Kreveld and M. Overmars, *Computational Geometry*, 3rd ed., §2.2. Areas use Green's theorem (J. Stewart, *Calculus*, 8th ed., §16.4).

11.9 Degree-of-freedom report

`sketch.dof(options)` checks a sketch as it is now (Section 4.6) and returns a `SketchDofReport`. Run it after `solve`. The lower-level `swx_sketch::dof::analyze(&sketch, sketch.params(), options)` returns the `DofReport` alone.

Setting	Meaning	Default
`rank_tolerance`	How independent an equation must be, compared with the largest, to count as a new rule rather than a repeat.	`1e-9`
`residual_tolerance`	A sketch with a repeated rule is `Conflicting`, not `OverConstrained`, when its remaining error is more than this.	`1e-8`

Report field	Meaning
`dof`	Degrees of freedom left: the sketch's numbers minus its independent equations.
`rank`	The number of independent equations.
`redundant`	The numbers of the equations that repeat earlier ones, smallest first, counting from 0 in the order the constraints were added.
`residual_norm`	The remaining error: how far the rules are from being met, overall.
`state`	One of the states below.

State	When
`UnderConstrained`	No repeated rule, and `dof` is more than 0.
`FullyConstrained`	No repeated rule, and `dof` is 0.
`OverConstrained`	At least one repeated rule, and every rule is met.
`Conflicting`	At least one repeated rule, and the remaining error is more than `residual_tolerance`.
`NonFinite`	A value is infinite or not a number. Nothing else was worked out.

SketchDofReport field	Meaning
`report`	The `DofReport` above.
`redundant_constraints`	The constraints that own a repeated equation, as `ConstraintId`s, smallest first, each listed once. Of two rules that repeat each other, the one added later is listed.

`sketch.constraint_of_row(n)` gives the constraint that made equation `n`, or `None` when there is no such equation.

Entity or constraint	Numbers or equations
Point · Line · Circle · Arc	2 · 4 · 3 · 5 numbers (degrees of freedom when free).
Coincident · Fixed · Midpoint · Symmetric	2 equations each.
Every other constraint	1 equation.

The count uses a Householder QR factorization, taking the equations in the order they were added; see G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., §5.2.

11.10 Inference

`swx_sketch::infer` suggests the constraints a newly added entity already almost satisfies (Section 5.8). `infer(&sketch, new, &options)` returns them and changes nothing. `add_inferred(&mut sketch, new, &options)` adds them in order, then solves, and returns the new constraint ids. Both fail if `new` doesn't exist or a setting is invalid.

Setting	Meaning	Default
`point_tol`	How close, in mm, two points must be to be joined, and how near a line or circle must come to touching to be made tangent. Finite and > 0.	diagonal ÷ 1000
`angle_tol_deg`	How close, in degrees, a line must be to level, upright, parallel or perpendicular. Finite and > 0.	1

`InferOptions::for_sketch(&sketch)` gives the defaults for that sketch. The diagonal is that of the smallest axis-aligned box holding the whole sketch, with circles and arcs counted as full circles; for an empty sketch `point_tol` is 0.001. The suggestions come in this order:

Suggestion	When
Coincident	For each point of the new entity (a line's start, then end; a circle's center; an arc's center, start, then end), the nearest other point within `point_tol`, if any: a point, a line end, an arc end or a center. Ties go to the entity added first.
Horizontal, vertical	The new entity is a line within `angle_tol_deg` of level or upright.
Parallel, perpendicular	The new line is within `angle_tol_deg` of parallel or square to another line. At most one, the closest. When the new line got horizontal or vertical, lines that are themselves level or upright are skipped.
Tangent	A line whose distance to a circle's or arc's center is within `point_tol` of the radius (the line counted as endless); or two circles or arcs whose centers are within `point_tol` of touching, outside or inside. Circles with the same center are skipped.

A constraint the sketch already holds, in either order, is never suggested, so inferring twice adds nothing. Inference never suggests a dimension.

Error from add_inferred	Meaning
`Sketch`	The entity doesn't exist, or a setting is not finite or not > 0. Nothing is added.
`Solve`	The solver could not start.
`NotConverged`	The constraints were added, but the solve did not finish; the geometry is the solver's best attempt.

11.11 Parameters

A `swx_core::Document` keeps its parameters in creation order (Section 10.3). Every call that can fail leaves the table unchanged when it does.

Call	What it does
`add_model_param(value, unit)`	Makes a model parameter named `d<N>` and returns the name. `N` starts at 0 and only goes up; a name already taken is skipped.
`add_user_param(name, value, unit)`	Makes a user parameter. Fails if the name is invalid or taken.
`rename_param(old, new)`	Renames a parameter, every feature that uses it, and every formula that names it (whole names only: renaming `d1` leaves `d10` alone). The kind is kept. Renaming to the same name does nothing.
`set_param_comment(name, comment)`	Sets the comment. Fails if there is no such parameter.
`set_param(name, value)`	Sets the value and drops the parameter's formula, if it had one, then works out every formula that depends on it. If one of those fails, they keep their old values, with no error. An unknown name makes a user parameter with an empty unit; this name is not checked.
`set_param_expr(name, text)`	Gives the parameter a formula (Section 11.12) and works out the whole table. The result is stored in the parameter's unit; a unitless result is taken in that unit. Fails if the parameter or a name in the formula is unknown, the formula doesn't parse or can't be worked out, its result is the wrong kind of quantity for the unit, or formulas would form a loop.
`evaluation_order()`	Every parameter name, each after the parameters its formula names; ties keep creation order.
`dirty_after_param_change(name)`	The features to rebuild when `name` changes, including those that use a parameter whose formula depends on it, directly or through other formulas.
`param(name)`	The value, or `None`.
`parameters()`	Every parameter, in creation order.
`is_valid_param_name(name)`	Whether a name is allowed.

Parameter field	Meaning	Allowed values
`name`	Unique within the part. Case-sensitive.	a letter or `_`, then letters, digits or `_` (ASCII)
`kind`	Where it came from.	`Model`, `User`
`value`	The number; for a parameter with a formula, its latest result.	any number
`unit`	Free text; not checked. Formulas understand only the units in Section 11.12.	any, may be empty
`comment`	A note for people.	any; empty by default
`expr`	The formula as typed, or `None` for a plain number.	`None` by default

The dependency order uses Kahn's algorithm, always taking the earliest-made parameter that is ready; see A. B. Kahn, "Topological sorting of large networks", *Communications of the ACM* 5(11), 1962, and T. H. Cormen et al., *Introduction to Algorithms*, 3rd ed., §22.4.

A unit that is not one of the seven in Section 11.12, including an empty one, counts as unitless when a formula reads or stores the parameter.

`swx_sketch::params` ties a sketch's dimensions to a document's parameters (Section 10.4). A sketch remembers each binding as a pair of constraint and parameter name; `sketch.bindings()` lists them in the order they were made.

Call	What it does
`bind_dimension(sketch, doc, c)`	Adds dimension `c` (a `Distance`, `Radius` or `Diameter`) to the sketch, and a model parameter with the same value and unit `mm` to the document; returns the constraint id and the parameter's name. A diameter's parameter holds the diameter. On error neither changes.
`apply_params(sketch, doc)`	Copies each bound parameter's current value into its dimension. Checks every value first, so on error nothing changes. Solve afterwards to move the geometry.
`rename_bound(sketch, old, new)`	Points the sketch's bindings from `old` to `new`; returns how many changed. Call it with every `doc.rename_param`.
`sketch.set_dimension_value(id, value)`	Changes one dimension's value directly. Value finite and > 0.

11.12 Expressions

`swx_core::expr` reads and works out expressions (Section 10.5). `parse(text)` turns the text into an `Expr`; `eval(&expr, &resolve, default_length)` works it out, asking `resolve(name)` for each name's value. `expr.names()` lists the names an expression uses, each once, in first-use order. The result is a `Quantity`: a `value` in millimeters and radians and a `dim` (its powers of length and angle). `q.in_unit(unit)` converts the value for display. `default_length` is the unit for showing lengths; it doesn't change the result.

Part	Rule
Numbers	`5`, `2.5`, `.5`, `1e-3`. A bare number has no unit.
Names	A letter or `_`, then letters, digits or `_`, as for parameters.
Operators	`+ - * /`, unary `-`, and `( )`. `*` and `/` before `+` and `-`; left to right otherwise.
Units	Written straight after a number or a `)`: `5 mm`, `(2 + 3) mm`. After a name, a word is not a unit.
`+` and `-`	Both sides must have the same dimension: two lengths, two angles, or two bare numbers.
`*` and `/`	Any dimensions; they multiply or divide. A length times a length is an area.
Limits	At most 1024 pieces and 64 levels of brackets or minus signs.

Unit	Measures	Size
`mm`	length	1 mm
`cm`	length	10 mm
`m`	length	1000 mm
`in`	length	25.4 mm
`ft`	length	304.8 mm
`deg`	angle	π/180 rad
`rad`	angle	1 rad

11.13 File formats and units

Format	Read / write	Notes
STEP (`.step` , `.stp`), ISO 10303-21	Both	Solids only. Written in millimeters.
PNG (`.png`)	Write	Pictures from `swx-qa render` and `swx-qa sketch` .
Saulitwerx part (`.sltwx`)	Not yet	Planned: the part with its full history.

Quantity	Unit
Length	millimeter (mm)
Area, volume	mm^2, mm^3
Angle	degrees for feature parameters such as a revolve's sweep; radians for arc angles in sketches and for the geometry engine's `revolve` and `revolve_feature`

11.14 Keyboard and mouse

There are no keyboard or mouse controls yet, because there is no app window. They arrive with the desktop app and will be listed here.

PART V

Help

What to do when something goes wrong: symptoms, every error message you can see, common questions and known limitations.

CHAPTER 12

Help

12.1 Troubleshooting

Symptom	What to do
`cargo: command not found`	Rust isn't installed, or this terminal started before it was. Open a new terminal; if that doesn't help, repeat the Rust step in Chapter 2.
`could not find Cargo.toml`	The terminal isn't in the `saulitwerx` folder. Run `cd saulitwerx` and try again.
A build with `--features occt` fails with errors about `.hxx` files or `TK…` libraries	OpenCascade isn't installed. Install the `libocct-…-dev` packages from Chapter 2.
`swx-qa: cannot read … STEP input needs OCCT`	`swx-qa` was built without OpenCascade. Rebuild it: `cargo build --release -p swx-qa --features occt`.
`target/release/swx-qa: No such file or directory`	The tool isn't built yet. Run the build step in Chapter 2.
Rows of `*` and colored text appear when saving or reading STEP	Normal. The geometry engine prints its own report. Saulitwerx's own lines come before and after it.
The picture looks small or blurry when printed	Use a larger `--size`: about 80 pixels per centimeter of printed width.
A curved part is covered in lines	Raise `--crease`, for example to 45.
The numbers in a sketch picture are too small to read	Use a smaller `--size`. Labels have a fixed size in pixels, so they take up more of a small picture.
A sketch solves, but its sizes aren't what you expect	Some of it is still free to move. Check the degrees of freedom (Section 5.3) and add the missing dimensions.
A sketch looks closed, but has no region	Some ends don't meet. The open-edge list names the pieces (Section 5.7). Join every end, line or arc, to its neighbor with coincident (Section 5.6).
An arc flips or bulges the wrong way after solving	An arc runs counterclockwise from `ArcStart` to `ArcEnd`. Check that each line end is joined to the right end of the arc (Section 5.6).
A sketch dimension ignores a new parameter value	Copy the values into the sketch with `apply_params`, then solve again (Section 10.4).
Inference left a corner open	The ends were farther apart than `point_tol`. Draw the new piece from where the last one ended, or widen `point_tol` (Section 5.8).
Inference made a line level that you meant to lean	It was within `angle_tol_deg` of level. Lower the setting, or don't add the suggestion (Section 5.8).
A parameter stopped following its formula	`set_param` replaced the formula with a plain number. Give it the formula again with `set_param_expr` (Section 10.6).
A formula's value didn't change after its inputs did	The new inputs made the formula fail, for example by dividing by zero, so it kept its old value. Change the inputs back, or give a formula that works (Section 10.6).
The solver stops with `Stagnated`	Two or more rules contradict each other. Check the degrees of freedom: the report says `Conflicting` and gives the clashing equation's number.

12.2 Error messages

From `swx-qa`

Every message starts with `swx-qa:` . No picture is written when there is an error.

Message	Meaning and fix
`missing command`	Nothing after `swx-qa` . Add `render` , `sketch` , or `--help` .
`unknown command `x``	Only `render` , `sketch` and `help` exist.
`unknown option `x``	A misspelled option, or one that belongs to the other command (`--crease` and `--step` are for `render` only). The help text follows the message.
`` `--size` needs a value ``	An option was last on the line without its value. The same goes for the other options.
`missing input: give --fixture <name> or --step <file.step>`	Say what to draw.
`give exactly one of --fixture and --step`	Two inputs were given. Keep one.
`missing input: give --fixture <name>`	`sketch` needs the name of a built-in sketch.
`missing output: give -o <out.png>`	Say where to save the picture.
`unknown fixture `x`; available: cube, cylinder, l-bracket`	Use one of the listed names.
`unknown sketch fixture `x`; available: rectangle, triangle-345`	Use one of the listed names.
`invalid --size `x`: expected <W>x<H>, e.g. 800x600`	Write the size as two whole numbers joined by `x` , with no spaces.
`invalid --size `x`: width and height must be > 0`	Neither side may be 0.
`image size WxH must be between 1x1 and 8192x8192`	Use a smaller size.
`invalid --crease `x`: expected degrees`	Give a number, such as `45` .
`crease angle x must be between 0 and 180 degrees`	Use a value from 0 to 180.
`cannot read f: invalid input: no such file: f`	The STEP file isn't there. Check the name and the folder.
`cannot read f: kernel failure: import_step from f failed`	The file isn't valid STEP. The geometry engine prints the reason just above.
`cannot write f: …`	The picture can't be saved there, for example because the folder doesn't exist.
`mesh is empty` , `mesh has no visible extent`	The STEP file holds nothing that can be drawn.

From the geometry kernel

Message	Meaning and fix
`invalid input: <name> must be finite and > 0, got <v>`	A size, radius or height is zero, negative or not a number.
`invalid input: <name> must have finite coordinates, got …`	A position contains an infinite value or not-a-number.
`invalid input: no such file: f`	The STEP file to read doesn't exist.
`invalid input: adjacent needs two different kinds, got edge and edge`	`adjacent` was asked for the same kind it starts from. Ask for another kind: the faces or vertices of an edge (Section 6.3).
`invalid input: face index 7 is out of range: the shape has 7 faces`	The index is past the last sub-shape of its kind. Indices start at 0, so the last is the count minus 1. An index from another shape, or from before a change, can also be out of range (Section 6.3).
`kernel failure: <operation> failed`	OpenCascade couldn't complete the operation. For a cut, check that the tool solid overlaps the part (Chapter 6).

Message	Meaning and fix
`invalid input: outer loop: consecutive segments don't meet within 0.0000001` (or `hole N loop: …`)	Two pieces of the loop, or its last and first, don't meet. Reuse the same corner values for both ends (Chapter 7).
`invalid input: outer loop is empty`, `… loop encloses no area`	The loop has no pieces, or they enclose nothing: a line there and back, or a figure eight whose halves are the same size.
`invalid input: … loop of one segment must be a circle`, `… a circle must be the only segment of its loop`	A loop of one piece must be a full circle, and a circle can't share a loop. Use arcs to round corners.
`invalid input: line from … to … has no length`, `arc from … to … rad has no sweep; use a circle`	A piece starts and ends at the same point.
`invalid input: radius must be > 0, got <v>`	An arc or circle has a zero or negative radius.
`invalid input: line must be finite, got …` (also `arc`, `circle`, `workplane origin`)	A number is infinite or not a number.
`invalid input: workplane x_dir must be non-zero` (or `normal`), `workplane x_dir and normal must be perpendicular (dot …)`	The workplane's directions are missing or not at right angles (Section 11.5).
`kernel failure: make_face failed: the face is invalid (e.g. a self-intersecting loop or a hole outside the outer loop)`	The loops pass the checks above, but the outline crosses itself, or a hole is outside the outline or touches its edge.
`invalid input: extrude distance must be finite and non-zero, got <v>` (or `symmetric extrude length`)	The extent is zero or not a number.
`kernel failure: extrude failed`	The shape given to `extrude` or `extrude_tracked` isn't a face from `make_face` or `make_face_tracked`.
`invalid input: extrude mode "join" needs a target body` (or `"cut"`, `"intersect"`)	The mode combines with a part, but none was given. Pass the part as the target, or use `NewSolid` (Section 7.3).
`invalid input: a through-all extent works only with the cut mode, not "join"` (or another mode)	Only a cut can go through all. Give a distance instead (Section 7.4).
`invalid input: a through-all extent needs a target body to measure; use extrude_feature with the cut mode`	A through-all extent was given to `extrude` or `extrude_tracked`, which have no target to measure. Use `extrude_feature` with `ExtrudeMode::Cut`.
`invalid input: the through-all cut misses the target: it removes nothing`	The cut went the wrong way, or the outline isn't over the part. Change the direction or move the outline (Section 7.4).
`invalid input: revolve angle must be in (0, 2π] radians, got <v>`	The angle is zero, negative, more than a full turn, not a number, or given in degrees. Give radians: a full turn is 2π (Section 8.1).
`invalid input: revolve axis direction must be non-zero` (or `must be finite, got …`), `revolve axis origin must be finite, got …`	The axis is missing its direction, or has a number that isn't finite.
`invalid input: the profile crosses the revolve axis (origin …, direction …): it reaches <a> on one side and <b> on the other`	The axis runs through the outline. Move the axis or the outline so the outline is all on one side; touching the axis is fine (Section 8.3).
`invalid input: revolve axis (origin …, direction …) is not in the profile's plane (through …, normal …)`	The axis is tilted out of the outline's plane, stands straight out of it, or runs beside it. Use an axis that lies in the plane (Section 8.3).
`invalid input: revolve needs a single planar face`	The shape given to the revolve isn't one flat face. Pass a face from `make_face`.
`kernel failure: revolve failed`	The geometry engine couldn't make the revolve, although its input passed every check above.
`invalid input: revolve mode "join" needs a target body` (or `"cut"`, `"intersect"`)	As for an extrude: pass the part as the target, or use `NewSolid` (Section 8.2).

Message	Meaning and fix
`empty result: revolve new solid leaves no solid (volume …)` (or `join`, `cut`, `intersect`)	Nothing is left: for a cut or an intersect, as for an extrude.
`empty result: extrude cut leaves no solid (volume …)` (or `intersect`, `join`)	Nothing is left: the cut removed the whole part, or the intersect found no overlap. Make the outline smaller, or move it over the part.

From sketches

Message	Meaning and fix
`geometry values must be finite`	A coordinate, radius or angle is infinite or not a number.
`radius must be > 0, got r`	A circle or arc needs a positive radius.
`value must be > 0, got v`	A distance, radius or diameter constraint needs a positive value.
`unknown geometry N`	The constraint names an entity that isn't in this sketch.
`geometry N has a different kind`	Wrong kind of point or entity, such as `Center` of a line, `ArcStart` of a circle, a circle in parallel, or a line made equal to a circle (Chapter 5).
`unknown constraint N`	No constraint with that number is in this sketch.
`constraint N is not a dimension`	Only a distance, radius or diameter has a value to change.
`only distance, radius and diameter can be bound, got …`	`bind_dimension` was given another kind of constraint. Add it with `add_constraint` instead (Section 10.4).
`a sketch dimension is bound to unknown parameter `p``	The parameter was renamed or never existed. After `doc.rename_param`, call `rename_bound` with the same names (Section 10.4).

From profiles

Message	Meaning and fix
`geometry A and B intersect`	Pieces A and B cross, overlap, or touch away from their ends. Move or shorten one so they meet only end to end (Section 5.7).
`weld tolerance must be > 0, got t`	Give a `weld_tolerance` greater than zero, or `None` for the default.

Any sketch message above can also come from profiles, when the sketch itself is at fault.

From the document model

Message	Meaning and fix
`unknown parameter `p``	Set the parameter before a feature uses it. Renaming or commenting on a parameter that doesn't exist also gives this.
`a parameter named `p` already exists`	Every parameter needs its own name. Pick another, or rename the other parameter first (Section 10.3).
`invalid parameter name `p`: must start with a letter or `_` and contain only letters, digits and `_``	Use a name like `hole_2`: no leading digit, no spaces, dashes, dots or accents.
`unknown feature N`	No feature with that number is in the history.
`feature N references feature M which is not earlier in the history`	A feature can use only features made before it.
`expression for `p`: …`	The formula for `p` doesn't parse, names an unknown parameter, or can't be worked out. The rest is an expression message, below.
`parameter expressions form a cycle: a -> b -> a`	The formulas would depend on each other in a loop, traced in the message. Make one of them a plain number (Section 10.6).
`expression for `p` gives length^2, but its unit expects length`	The formula's result is the wrong kind of quantity for the parameter's unit, here an area for a length. A parameter with no unit, or a unit outside the seven, expects `unitless`.

From expressions

Each message gives the byte where the problem is, counting from 0 at the first character (Section 10.5).

Message	Meaning and fix
`syntax error at byte n: …`	The text isn't a formula: an unexpected character, a missing `)` or number, or an expression too long or too deeply nested. The rest of the message says which.
`unknown unit `u` at byte n (expected mm, cm, m, in, ft, deg or rad)`	A word after a number isn't one of the seven units. Check the spelling.
`unknown name `p` at byte n`	No parameter has that name. Names are case-sensitive.
`dimension mismatch at byte n: cannot apply `+` to length and angle`	Both sides of `+` or `-` must measure the same thing. Give a bare number its unit, as in `d1 + 5 mm`.
`division by zero at byte n`	The part after `/` works out to zero.

From inference

For an unknown entity, `infer` and `add_inferred` give the sketch messages above. They can also give:

Message	Meaning and fix
`geometry values must be finite`	A setting is infinite or not a number (Section 11.10).
`value must be > 0, got v`	A setting is zero or less. Both tolerances must be greater than zero.
`solve did not converge (residual r)`	From `add_inferred` only. The suggested constraints were added, but couldn't all be met. Check the degrees of freedom for a conflict (Section 5.3). A constraint can't be removed yet, so build the sketch again and add only the suggestions you want (Section 5.8).

12.3 Questions

Is there a version with windows and buttons? Not yet. The desktop app is a later milestone; see **Coming next**.

Can I open files from other CAD programs? Yes, if you save them as STEP first. Saulitwerx doesn't read any program's own native file format, and won't.

Can I save my design and edit it later? Not yet. Today you can save solids as STEP, which keeps the shape but not the history. Saulitwerx part files come later.

Does it cost anything? No. Saulitwerx is free and open source, under the MIT or Apache-2.0 license, at your choice.

Why millimeters? Mechanical design works mostly in millimeters, so it is the default. Choosing other units per part is planned.

12.4 Known limitations

- No desktop app, 3D view, mouse or keyboard controls yet.
- A sketch's regions can't be extruded or revolved directly yet: type the outline's pieces into a program (Chapter 7, Chapter 8). There is no fillet or hole yet.
- An extrude goes a set distance, or through all for a cut (Section 7.4); not "up to a face" or "between two faces".
- A revolve turns one way from the outline's plane by an angle; it can't yet turn both ways or symmetrically.
- An extrude's or a revolve's target is a solid you hold in your program. Faces and edges can be counted and numbered (Section 6.3), and a plain extrude says which face each piece of the outline became (Section 7.5), but they can't be picked, and their numbers don't last through a change; so the outline's workplane is given by its numbers (Section 7.3).
- Extrudes and revolves aren't recorded in a part's history, so they don't rebuild when a parameter changes.
- No native part files; work can't be saved with its history.
- `swx-qa sketch` draws only its two built-in sketches. Your own sketches need a short program (Section 9.5). Sketch pictures show no dimension arrows and no marks for constraints such as horizontal.
- A parameter's unit is free text. Only the seven units of Section 11.12 are understood by formulas; any other unit counts as none.
- When a new value breaks a formula that depends on it, the formula keeps its old value without an error (Section 10.6).
- A constraint can't be removed from a sketch once added; build the sketch again without it.
- Inference needs a program: there is no drawing with the mouse yet, so nothing is inferred as you move the pointer (Section 5.8).
- Renaming a parameter doesn't update the sketches bound to it by itself; call `rename_bound` too (Section 10.4).
- There are no angle dimensions and no point-to-line distance yet.
- Profiles don't split pieces where they cross; crossing pieces are an error.
- Only built on and tested on Ubuntu Linux.
- An imported STEP file becomes one shape, and can't be edited.

PART VI

Back matter

Glossary, what changed, what comes next, a one-page card, credits and licenses, and the index.

Glossary

Adjacent
Said of two sub-shapes when one bounds the other, such as a face and one of its edges.

Binding
The link between a sketch dimension and the parameter that drives it.

Axis of revolution
The line a revolve turns its outline around.

Bounding box
The smallest box, with sides parallel to the axes, that holds a solid.

Boolean operation
Combining two solids by union, cut or intersect.

CAD
Computer-aided design: using a computer to design physical objects.

Cap
One of the two flat faces at the ends of an extrude: where the push starts and where it ends.

Coincident
A constraint that puts two points at the same place.

Conflicting
Said of a sketch whose rules contradict each other, so no shape can meet them all.

Constraint
A rule about a sketch's shape, such as "horizontal" or "these points touch".

Cut
A Boolean operation that removes one solid from another.

Degree of freedom (DOF)
One way a piece of a sketch can still move. A free point has two.

Dependency order
The order formulas are worked out in: each parameter after the parameters its formula uses.

Diameter
The width of a circle through its center; twice the radius.

Dimension
A constraint with a number: a distance, radius or diameter.

Dirty
Said of a feature that must be rebuilt after a change.

Edge
A curve where two faces of a solid meet, or where one round face meets itself (a seam).

Equal
A constraint that makes two lines the same length, or two circles or arcs the same size.

Entity
One piece of sketch geometry: a point, line, circle or arc.

Extent
How far an extrude goes: a distance one way, or a length split evenly on both sides of the workplane.

Extrude
Pushing a flat outline straight out into a solid with straight sides.

Extrude mode
What an extrude does with its shape: keep it as a new solid, or join, cut or intersect it with a target body.

Face
One flat or curved patch of a solid's surface.

Expression
A small formula with units that works out a value, such as `d1 / 2 + 5 mm`.

Feature
One step in building a part, such as a sketch or a cut. Features are kept in the history.

Feature tree
Another name for a part's history.

Fixed
A constraint that keeps a point at exact coordinates.

Fully constrained
A sketch whose constraints and dimensions leave nothing free to move.

Geometry kernel
The part of a CAD program that does the 3D math. Saulitwerx uses OpenCascade.

History
A part's features, in the order they were made.

Inference
Saulitwerx adding the constraints a newly drawn piece already almost satisfies, such as coincident or horizontal.

Intersect
A Boolean operation that keeps only what two solids share.

Island
A region inside a hole of another region.

Isometric
A view from a corner, looking down at an angle, so that three sides of a box show equally.

Loop
A closed chain of sketch pieces, each ending where the next begins.

Midpoint
A constraint that keeps a point halfway along a line.

Model parameter
A parameter that belongs to a dimension, named `d0`, `d1`, … in turn.

Open edge
A sketch piece that isn't part of any closed loop.

OpenCascade (OCCT)
The open-source geometry kernel Saulitwerx uses.

Over-constrained
Said of a sketch with a rule that repeats what other rules already say.

Parallel
A constraint that makes two lines point the same way.

Parameter
A named number that features and dimensions refer to.

Parametric
Built from steps and numbers that can be changed later, with the design rebuilding to match.

Perpendicular
A constraint that makes two lines meet at a right angle.

Pre-alpha
Early, unfinished software: the parts work, but it isn't ready for real use.

Profile
The regions of a sketch: the areas a feature can turn into a solid.

Redundant
Said of a rule that repeats what other rules already say.

Region
A flat area enclosed by an outer loop, minus any holes inside it.

Revolve
Turning a flat outline around an axis into a round solid.

Revolve angle
How far a revolve turns: a full turn (2π radians, 360°) or part of one.

Seam
An edge where a closed round face, such as the wall of a hole, meets itself.

Side face
A face of an extrude swept by one edge of its outline: a flat wall from a line, a round one from an arc or circle.

Sketch
A 2D drawing on a plane, made of points, lines, circles and arcs.

Solid
A closed 3D shape with an inside and an outside.

Solver
The calculation that moves sketch geometry until every constraint holds.

Sub-shape
One face, edge or vertex of a solid.

Symmetric
A constraint that makes two points mirror images across a line.

STEP
An international standard file format (ISO 10303) for exchanging 3D parts.

Surface area
The total area of a solid's faces.

Tangent
Touching without cutting in: a line and a circle, or two circles, that meet at one point.

Target body
The solid an extrude or a revolve joins to, cuts from or intersects with.

Through all
An extrude cut with no depth of its own, which goes right through the target body.

Under-constrained
Said of a sketch with at least one degree of freedom left.

Union
A Boolean operation that joins two solids into one.

Unit
What a number measures in, such as millimeters.

User parameter
A parameter you create yourself, with a name you choose.

Vertex
A corner point of a solid, where its edges meet. Plural: vertices.

Volume
How much space a solid fills.

Workplane
The flat plane an outline is drawn on, with its own directions and a normal pointing straight out of it.

What’s new

2026-11-04: which face is which

- **New:** extrude with history, and find out which face of the solid each piece of your outline became: the two caps and every side face (Section 7.5, Section 11.5).
- **New:** ask for the bounding box of a single face, edge or vertex (Section 11.4).
- **Pictures:** the extruded plate, with its faces named by their history.
- Help: two new common mistakes; two error messages and a known limitation updated. Glossary: cap, side face.

2026-11-03: safer revolves

- **New:** a revolve now checks its axis first. An axis that runs through the outline, or doesn’t lie in the outline’s plane, gets a clear message saying which, instead of a broken solid or a vague failure (Section 8.3).
- The revolve sample shows both new errors (Section 8.1).
- Help: three new error messages; the revolve rows and the known limitation about unchecked axes updated.

2026-11-02: faces, edges and vertices

- **New:** count a solid’s faces, edges and vertices, and list which ones touch: the edges around a face, the faces along an edge (Section 6.3, Section 11.4).
- Concepts: vertices join faces and edges in Section 4.1, with the plate’s figure now marking one of each.
- Help: two new error messages; the known limitation about faces and edges updated.

2026-11-01: revolve, and cuts through all

- **New:** revolve a flat outline around an axis into a round solid, by a full turn or part of one, as a new solid or joined to, cut from or intersected with a part (Chapter 8, Section 11.5).
- **New:** an extrude cut can go right through a part, one way or both, however thick the part is (Section 7.4).
- **Pictures:** a flanged bushing at three quarters of a turn and at a full turn, the bushing with a groove, and a block with two through holes.
- Help: nine new error messages, eight new common mistakes, and known limitations updated. The limitations that there is no revolve and no “through all” are gone.

2026-10-31: extrude modes

- **New:** an extrude can join its shape to a part, cut it out, or keep only the overlap, in one step (Section 7.3, Section 11.5).
- **Pictures:** one block with a boss, a pocket and a trimmed shape made from it.
- Help: two new error messages, three new common mistakes, and known limitations updated. The limitation that an extrude always makes a new solid is gone.

2026-10-30: extrude

- **New:** extrude a flat outline into a solid. Outlines are made of lines, arcs and circles, can have holes, and can sit on any plane; the extrude goes a distance one way or splits evenly on both sides (Chapter 7, Section 11.5).
- **Pictures:** a profile and the plate extruded from it, and an extruded slot.
- Concepts: the profile and feature sections now point to the extrude.
- Help: eleven new error messages, and known limitations updated. The limitation that nothing turns outlines into solids is gone.

2026-10-29: formulas in parameters, and inference

- **New:** a parameter can hold a formula, such as `d0 / 2 + 5 mm`, and follows the parameters it names. Formulas are worked out in dependency order, loops and wrong units are refused, and renaming a parameter rewrites the formulas that use it (Section 10.6, Section 11.11).
- **New:** inference adds the coincident, horizontal, vertical, parallel, perpendicular and tangent constraints a newly drawn piece already almost satisfies (Section 5.8, Section 11.10).
- **Pictures:** a rough rectangle and circle before and after inference.
- Help: new error messages for formulas and inference, four troubleshooting rows, new common mistakes, and known limitations updated. The limitation that parameters hold only plain numbers is gone.

2026-10-28: dimensions as parameters, expressions, and joined arcs

- **New:** constraints can name an arc's two ends (`ArcStart`, `ArcEnd`), so lines and arcs join into closed outlines. The slot task is rewritten: it now starts from a careless drawing and solves to a fully defined slot (Section 5.6, Section 11.6).
- **New:** a sketch's distance, radius and diameter dimensions can each make a model parameter, `d0`, `d1`, …, and follow it when it changes (Section 10.4).
- **New:** work out expressions with units, such as `d1 / 2 + 5 mm`, with unit checking (Section 10.5, Section 11.12).
- **Pictures:** the slot before and after solving replaces the slot with a gap.
- Help: new error messages for bound dimensions and expressions; troubleshooting and known limitations updated. The limitation on arc ends is gone.

2026-10-26: parameter table, and named conflicts

- **New:** a part's parameters form a table: model parameters named `d0`, `d1`, … in turn, user parameters with names you choose, units and comments. Renaming a parameter updates every feature that uses it (Section 4.8, Section 10.3, Section 11.11).
- **Changed:** the degree-of-freedom report now names the redundant or conflicting constraints, not just equation numbers (Section 5.3, Section 11.9). The matching known limitation is gone.
- Help: two new parameter error messages, and two new common mistakes.

2026-10-25: more constraints, and profiles

- **New:** six constraints: parallel, perpendicular, equal, midpoint, symmetric and tangent (Section 4.4, Section 11.6).
- **New:** keep a hole centered whatever the plate's size (Section 5.5), and draw a slot with tangent, equal-sized ends (Section 5.6).
- **New:** find the regions a sketch encloses, with their holes and areas, and the pieces left open (Section 4.7, Section 5.7, Section 11.8).
- **Pictures:** the centered plate at two widths, and a slot with a gap, ringed by the template.

- Help: profile error messages, two new troubleshooting rows, and known limitations updated, including arcs whose ends can't be constrained.
- Corrected: midpoint and symmetric add two equations each to the degree-of-freedom count.

2026-10-24: degrees of freedom and sketch pictures

- **New:** check how fully a sketch is defined: degrees of freedom left, and whether it is under-, fully or over-constrained, or conflicting (Section 4.6, Section 5.3, Section 11.9).
- **New:** draw a sketch to a PNG with `swx-qa sketch` (Section 9.5, Section 11.2).
- **Pictures:** the sketch sections of **Concepts** and **Sketching** now show the plate sketch, rough and solved.
- Help: new troubleshooting rows and `swx-qa sketch` error messages; known limitations updated.

2026-10-22: first edition

The first edition of this manual. It covers everything on `main` today:

- Installing from source, and a first tutorial that builds, measures, saves and draws a plate with a hole.
- The ideas of parametric CAD: solids, units, sketches, constraints, dimensions, parameters, features and history.
- Sketching with the coincident, horizontal, vertical, fixed, distance, radius and diameter constraints, and the solver.
- Making and combining solids, and saving and reading STEP files.
- Drawing parts with `swx-qa render`.
- The feature history, and finding what a change rebuilds.
- Reference tables, every error message, troubleshooting, and known limitations.

Coming next

These are the next items on the public roadmap. None of them works yet. When one arrives, it moves into the chapters of this manual and into **What's new**.

Soon	What you'll be able to do
Extrude from a sketch	Hand a sketch's region straight to the extrude and the revolve, and extrude up to a face or between two faces.
More part features	Fillet, chamfer, shell and hole, driven by sketches.
Lasting references	A feature built on a face or an edge keeps finding it when earlier features change.
Part files	Save a part with its full history, and open it again.
Desktop app	A window with a 3D view, a model browser, sketch mode and property panels.

Later milestones add assemblies, 2D drawings and exploded views. The full plan is in `docs/ROADMAP.md` in the source code.

Quick reference

BUILD

TERMINAL

```
cargo test --workspace
cargo build --release -p swx-qa \
  --features occt
```

DRAW A PART

TERMINAL

```
swx-qa render --step part.step -o part.png
swx-qa render --fixture cube -o cube.png
  --size 1600x1000   (default 800x600)
  --crease 45     (default 30)
```

DRAW A SKETCH

TERMINAL

```
swx-qa sketch --fixture rectangle -o r.png
  (or triangle-345)
```

SAMPLES

TERMINAL

```
cargo run -p swx-sketch \
  --example plate_sketch
cargo run -p swx-sketch \
  --example sketch_dof
cargo run -p swx-sketch \
  --example centered_plate -- 60 25
cargo run -p swx-sketch \
  --example slot_sketch
cargo run -p swx-kernel --features occt \
  --example plate_with_hole -- plate.step
cargo run -p swx-kernel --features occt \
  --example extrude_plate
cargo run -p swx-kernel --features occt \
  --example revolve_bushing -- revolve
cargo run -p swx-core \
  --example feature_history
cargo run -p swx-core \
  --example parameter_table
cargo run -p swx-sketch \
  --example driven_plate
cargo run -p swx-core \
  --example expressions
cargo run -p swx-core \
  --example parameter_formulas
cargo run -p swx-sketch \
  --example inferred_sketch
```

SKETCH CONSTRAINTS

Constraint	Keeps
Coincident	two points together
Horizontal	a line level
Vertical	a line upright
Fixed	a point at (x, y)
Distance	two points a length apart
Radius / Diameter	a circle's size
Parallel · Perpendicular	two lines' directions
Equal	two lengths or radii alike
Midpoint	a point mid-line
Symmetric	two points mirrored
Tangent	a line touching a curve

SKETCH STATES

State	Means
Under-constrained	can still move: add rules
Fully constrained	done: 0 DOF left
Over-constrained	a rule repeats others
Conflicting	rules contradict

SOLIDS

Call	Gives
`make_box`	box from corner + sizes
`make_cylinder`	cylinder on +Z
`make_sphere`	sphere
`union` · `cut` · `intersect`	Boolean result
`make_face` · `extrude`	outline ⟶ face ⟶ solid
`extrude_feature`	extrude and combine
`revolve_feature`	revolve and combine
`export_step` · `import_step`	STEP file out / in

UNITS

Millimeters. Angles: degrees in feature parameters; radians in sketch arcs and kernel revolves.

Credits and licenses

Saulitwerx is written by its contributors and is licensed under either the MIT license or the Apache License 2.0, at your choice. It is an independent project, not affiliated with or derived from any commercial CAD product.

Component	License	Notes
OpenCascade Technology (OCCT) 7.6	LGPL-2.1 with the OCCT exception	The geometry kernel. Used as a separate, dynamically linked library; not included in Saulitwerx's source.
`png`, `flate2`, `miniz_oxide`, `fdeflate`, `crc32fast`, `adler2`, `simd-adler32`, `bitflags`, `cfg-if`	MIT, Apache-2.0, Zlib or 0BSD	Write the PNG pictures of `swx-qa`.
`cc`, `shlex`, `find-msvc-tools`	MIT or Apache-2.0	Build the bridge to OpenCascade.
Typst 0.15.1	Apache-2.0	Typesets this manual.
Inter 4.0, by The Inter Project Authors	SIL Open Font License 1.1	Headings and labels in this manual. License in `docs/manual/fonts/OFL-Inter.txt`.
Libertinus Serif, DejaVu Sans Mono	SIL Open Font License 1.1; Bitstream Vera license	Body text and code in this manual, as bundled with Typst.

Every picture of a part in this manual was drawn by Saulitwerx's own renderer, `swx-qa render`, and every picture of a sketch by `swx-qa sketch`'s plotter, from geometry made by Saulitwerx.

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